The Big Pool Bangs Theory

The Big Pool Bangs Theory: Cosmology via Jammed Phase Transitions and the Hard-Sphere Kinetic Bounce

Abstract. We explore a mechanical alternative to inflationary ΛCDM in which the universe begins as a jammed ensemble of $\sim 10^{60}$ Planck-mass relics, treated as effective hard spheres (an ansatz motivated by the Compton–Schwarzschild coincidence). A loop-quantum-cosmology-type modified Friedmann equation, with critical density conjecturally identified with the hard-sphere jamming density, produces a non-singular bounce; the subsequent chaotic rebound ejects most of the ensemble. The transition is argued to begin as many small nucleation bubbles whose merger history seeds the rebound turbulence — a cascade nucleation model reproduces the needed kick statistics at order unity, pending an end-to-end run — and toy N-body simulations with the (independently motivated) turbulent kick distribution eject $87.9\% \pm 3.5\%$ of the mass — consistent with the observed dark-matter fraction, though the identification requires an unspecified relic-to-plasma conversion channel for the trapped remainder — and exhibit clean velocity sorting of the ejecta into radial shells. We confront the framework with data: the coasting expansion it naively implies is disfavored by Pantheon+ ($\Delta\chi^2 \approx 84$) and DES-SN5YR ($\Delta\chi^2 \approx 131$) relative to ΛCDM; a toy effective $w(z)$ from the measured shells evolves, as the framework requires, but with the opposite drift sign to the DES+DESI $w_0 w_a$ best fit, and a fully relativistic LTB lightcone calculation yields mild deceleration ($q_0^{\text{eff}} = +0.16$) rather than the observed acceleration; and a Planck 2018 radial-profile test of a predicted Cold Spot hot ring at 5ř–7ř returns a null. We organize the surviving claims into four hypotheses of decreasing confidence, catalogue seven falsifiable predictions with the instruments required to test them, and state openly the unsolved problems — causal structure of the bounce, relativistic formulation of ejection, the horizon and flatness problems, and the expansion history — that must be resolved for the core hypothesis to constitute a physical theory.

1. Introduction: Vacuum Crystallization & The Hard-Sphere State

Standard Big Bang cosmology (ΛCDM) relies on exponential scalar-field inflation (e^{Ht}) to explain why the observable universe is spatially flat, isotropic, and thermally uniform. However, inflation introduces severe fine-tuning constraints and singularity theorems (e.g., the Borde-Guth-Vilenkin theorem).

The **Big Pool Bangs Theory** presents an alternative, mechanical framework grounded in non-linear phase-transition mechanics and quantum fluid dynamics. Space begins in a false vacuum state carrying an immense energy density near the Planck limit ($\rho_{\text{vac}} \sim \rho_P \approx 10^{96}$ kg/m^3). Quantum fluctuations trigger a **vacuum crystallization phase transition** — and, as in any first-order transition, it does not begin as one large event: crystallization nucleates at *many* small points throughout the region, little bubbles of converted phase that grow, collide, and merge. The "bang" is the percolation of these crystallites into a single jammed state. (Timeline caveat: walls propagate at $\leq c$, so percolating a $\sim 10^{20}\,\ell_P$ patch takes at least $\sim 10^{20}\,t_P \approx 5 \times 10^{-23}$ s divided by the number of nucleation sites per crossing length — the nucleation density is a free parameter of the framework, and any density sparse enough to give nontrivial merger statistics implies a jamming assembly time of many orders of magnitude above t_P. Earlier drafts' "fractions of a Planck time" language is retracted; the assembly timescale and its consistency with trapped-surface avoidance in Sec. 3 are open, per Sec. 10 item 6.) This multi-bubble origin is not incidental: it removes the fine-tuned question of why exactly one nucleus would form, and the randomness of the merger history is inherited by the jammed core as internal turbulence — potentially supplying, as a fossil of the little bubbles, the stochastic kick spectrum that the rebound dynamics of Sections 3 and 9.1 otherwise require as an unexplained input — though whether nucleation-era velocity structure survives the collisional jammed epoch (where mean free paths are far below a particle radius and perturbations tend to thermalize) rather than being erased before the slingshot phase is an open question; at minimum the nucleation spectrum sets the *initial condition* that the bounce then reprocesses. We tested this quantitatively (Sec. 9.1): a nucleation-and-merger simulation (Poisson nucleation in a ball, growing walls, vector-summed wall impulses) recovers the Maxwellian form of the kick distribution (speed-distribution coefficient of variation 0.41–0.42, the 3D Maxwellian value, across all nucleation rates). We flag honestly that this is largely a central-limit-theorem consequence of summing many independent wall impulses rather than transition physics — a consistency check on the ansatz, not a derivation; the nontrivial statistic is the coherent-to-dispersive ratio $\sigma_{\rm rel}$. The merger statistics alone are turbulence-dominated ($\sigma/\langle v_r\rangle \approx 5$–$8$): nucleation supplies the dispersion, while the coherent radial component $\bar{f}$ must come from the bounce itself; the requirement $\sigma_{\rm rel} \approx 1$ thus translates into a demand for a coherent radial component the Poisson statistics cannot supply. **A cascade nucleation model supplies it.** If, instead of independent Poisson nucleation, the transition begins with a *single seed bubble whose wall catalyzes further nucleation* — a chain reaction propagating outward, halted by a jamming feedback condition at $\eta \approx 0.64$ — the conversion front sweeps radially from the seed and the wall impulses acquire a coherent outward bias. Simulating this cascade (wall-area-weighted triggering just outside existing bubbles, five seeds per parameter set) yields $\sigma_{\rm rel} = 0.65$–0.78, robust across a $4\times$ range in trigger distance — compared to 5–8 for Poisson and the ≈ 0.75–1.0 window the observed dark-matter fraction requires. The cascade's speed distribution is broader than Maxwellian (coefficient of variation 0.58–0.67 vs 0.42; the correlated, ordered front breaks the central-limit conditions), and its $\sigma_{\rm rel} = 0.65$–0.78 overlaps the required 0.75–1.0 window only at the edge. Three honesty notes gate the closure claim. First, the chain "cascade $\rightarrow$ kick spectrum $\rightarrow$ 85% ejection" is at present a *splice*: the 87.9%$\pm$3.5% ejection figure was produced by the Maxwellian-ansatz N-body runs, and the cascade's own kick field (heavier-tailed, radially ordered) has not yet been fed through the N-body pipeline end-to-end. Second, "no fitted kick parameters" means no *per-particle* fitting — the model retains structural choices (trigger distance, scanned $4\times$; nucleation lag; wall-area weighting; the jamming arrest $\eta = 0.64$; and the impulse-collection window dt_w, whose sensitivity we have not scanned and on which $\sigma_{\rm rel}$ may depend). Third, the kick recipe assumes a universal wall-momentum efficiency; in first-order-transition physics the bulk-fluid kinetic efficiency is model-dependent and typically small (Espinosa–Konstandin–No–Servant). The chain therefore *plausibly* closes at order unity, pending an end-to-end run and a window-sensitivity scan. We also note the cascade, being seeded at a point, reintroduces a preferred center — the isotropy diagnostics of the earlier ensemble were run on Poisson-style kicks only, so Sec. 6.1's "no mechanism for a preferred axis" statement does not yet cover the cascade origin, which may in fact provide one (a computation worth doing, given Hypothesis 2).

Because quantum geometry prohibits localized infinite densities, field gradients cannot collapse into point singularities. Instead, energy "crystallizes" into an ensemble of $N \approx 10^{60}$ irreducible, incompressible **Planck-mass relics** ($m_P = \sqrt{\hbar c/G} \approx 21.8\,\mu$g). These hard-sphere relics saturate local volume up to the random close-packing limit ($\eta \approx 0.64$), forming a jammed nucleus of finite radius:

$$R_{\text{jam}} = \left(\frac{3M_{\text{total}}}{4\pi\rho_P\eta}\right)^{1/3} \approx 3.1 \times 10^{-15} \text{ meters}$$

(The numerical value is convention-dependent at the factor-few level: with $\rho_P = m_P/\ell_P^3$ one obtains $R_{\text{jam}} \approx 1.2 \times 10^{-15}$ m. We also note $M_{\text{total}} = Nm_P \approx 2.2 \times 10^{52}$ kg is $\sim 7\times$ below the observable universe's matter mass $\sim 1.5 \times 10^{53}$ kg; $N \approx 10^{60}$ should be read as order-of-magnitude.)

2. The Quantum Duality of Planck-Mass Spheres

How can these relics behave as classical "hard spheres" rather than quantum wave-functions? At the Planck scale, quantum mechanical wave-packet spreading and general relativistic gravitational collapse reach an exact mathematical intersection known as **Compton-Schwarzschild Duality**:

1. **Quantum Spatial Uncertainty:** For a particle of mass m, its quantum spatial extent is bounded by its reduced Compton wavelength ($\lambda_C = \hbar/mc$).

2. **Classical Gravitational Radius:** The event horizon for mass m is its Schwarzschild radius ($R_s = 2Gm/c^2$).

3. **The Planck Scale Intersection:** Setting $\lambda_C = R_s$ yields exactly the Planck mass ($m \approx m_P$). At this unique mass:

$$\lambda_C \approx R_s \approx \ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35} \text{ m}$$

The Planck mass thus marks the regime where quantum wave-packet spreading and gravitational self-confinement are of the same order, and where neither quantum field theory nor classical general relativity is separately valid. **We therefore adopt, as a modeling ansatz rather than a derivation, the assumption that Planck relics possess a minimum spatial extent $\sim \ell_P$ that no interaction can compress**, and treat them as effective incompressible hard spheres. This assumption is falsifiable through the phenomenology it generates (Sections 6–8), and its microphysical justification is left as an open problem (Section 10).

3. The 3-Body Kinetic Bounce and the "Cosmic Billiards" Break

How can a core of mass $M \approx 10^{52}$ kg compress into a radius of 3.1×10^{-15} m (roughly the size of a proton) without instantly collapsing into a singularity?
The misconception arises from applying the static Schwarzschild metric to a dense fluid. Inside a uniform fluid, the space-time is described by the **FLRW metric** and its dynamics are governed by the **Raychaudhuri Acceleration Equation**:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right)$$

Near random close packing ($\eta \to \eta_{\text{rcp}} \approx 0.64$), the hard-sphere kinetic pressure diverges according to the free-volume (Salsburg–Wood) equation of state, $P \propto nk_BT\,(1 - (\eta/\eta_{\text{rcp}})^{1/3})^{-1}$. Crucially, however, in classical general relativity this **positive** pressure *gravitates*: a divergent P makes $(\rho + 3P/c^2)$ more positive and accelerates collapse rather than halting it. No classical equation of state with $P > 0$ can bounce an FLRW collapse. The rebound must therefore be sourced by quantum-geometric corrections that become dominant precisely when the relic ensemble reaches the jamming density $\rho_c = \eta_{\rm rcp}\,\rho_P$. We model this with a modified Friedmann equation of the form established in loop quantum cosmology:

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right), \qquad \rho_c = \eta_{\rm rcp}\,\rho_P$$

The correction term $(1 - \rho/\rho_c)$ encodes the same physics as the hard-sphere picture — a maximum attainable density set by the incompressibility of Planck relics — but does so covariantly: $H \to 0$ as $\rho \to \rho_c$, the contraction halts, and the effective Raychaudhuri equation flips sign, driving an elastic rebound. In this framework the jamming limit of the relic gas is proposed as the physical origin of the critical density ρ_c that loop quantum cosmology introduces axiomatically. We note the honest comparison: LQC's own Immirzi-fixed value is $\rho_c \approx 0.41\,\rho_P$, versus $\eta_{\rm rcp}\rho_P = 0.64\,\rho_P$ here — agreement at the order-of-magnitude level only, and the identification should be read as a coincidence to be explained, not a derivation. Fig. 1 illustrates the two dynamical regimes; the bounce at ρ_c is built into the modified equation, not independently confirmed by it. A numerical integration of both cases (Fig. 1) confirms that the classical hard-sphere fluid undergoes runaway collapse through the jamming density, while the modified equation produces a non-singular bounce exactly at $\rho = \rho_c$. If the bounce completes before a trapped surface forms around the still-homogeneous core, the interior evolves as a bouncing FLRW patch rather than a black hole — but the multi-bubble assembly timeline of Sec. 1 makes the bounce epoch itself uncertain, so this ordering is an assumption, not a result; a rigorous trapped-surface timing analysis is deferred to future work (Section 10).

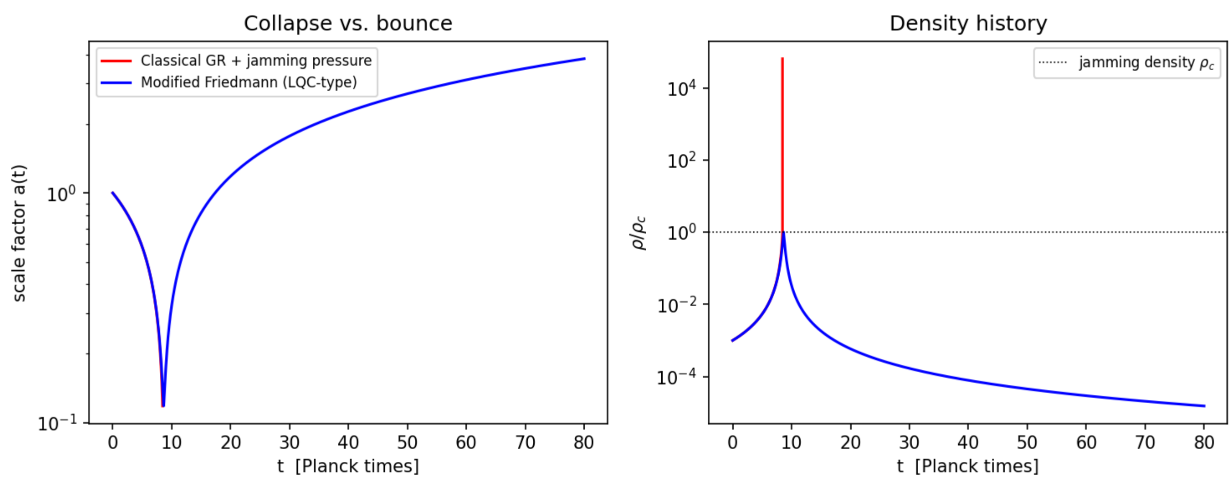

Figure 1: Fig. 1 — Toy FLRW integration. Left: scale factor. Right: density history. Classical GR with jamming pressure (red) overshoots the jamming density by $> 10^4$ (collapse); the modified Friedmann equation (blue) bounces at ρ_c.

But this is not a smooth, uniform expansion. It triggers the most chaotic event in physics: a 10^{60}**-body gravitational slingshot**. *A note on regime before proceeding:* everything that follows in this section, and all simulations in Sec. 9.1, are **Newtonian analogies** — deliberate proof-of-principle

caricatures of dynamics that in reality occur in a regime where escape velocities formally exceed c by ~ 20 orders of magnitude and only a general-relativistic treatment (ejecta as streams on an expanding background) is meaningful. We use the Newtonian model because violent-relaxation statistics (which fraction unbinds, how ejecta sort by velocity) are plausibly governed by dimensionless energy ratios that survive the translation; whether they actually do is the central open problem of Sec. 10 item 6.
Imagine the primordial universe not as a magical singularity, but as an unimaginably dense, microscopic game of three-dimensional billiards. Right before the Big Bang, space crystallizes into 10^{60} fundamental billiard balls (the Planck relics). Packed to the "jamming limit," they literally cannot compress any further. However, possessing immense mass and gravitational pull, they are desperately trying to fall inward. The pressure builds toward infinity.
When the quantum-geometric bounce reverses the collapse and the core rebounds, the ensuing expansion is highly turbulent. To understand how this ejects matter, we must look at the classical 3-body problem. When two bodies interact in isolation, their gravity creates a peaceful, predictable orbit. But the moment you add a third body of similar mass, the mathematics of gravity becomes completely chaotic.
The three objects do not form a stable system. Instead, they violently yank and pull on each other until a specific physical event occurs: the **gravitational slingshot**. In almost every 3-body interaction, two of the bodies will eventually fall closer together, dropping into a tighter, lower-energy orbit. By dropping deeper into each other's gravity well, they release a massive amount of kinetic energy. This energy does not vanish; it is instantaneously transferred to the third body. The third body absorbs this stolen energy and is violently kicked out of the system at extreme speeds.
At the jamming density itself the relic gas is collisional and hydrodynamic — mean free paths are far shorter than a particle radius, and no ballistic slingshot can operate. The slingshot phase begins *after* the bounce, once expansion dilutes the ensemble below the contact (connectivity) threshold and relics decouple onto ballistic trajectories. From that point onward the system is a dense, self-gravitating N-body ensemble, and multi-body slingshots dominate its relaxation. The balls smash into each other, and every time a cluster gets tangled up, the slingshot effect kicks a portion of them outward at blinding speeds.
In astrophysics, when Newtonian computer simulations are run on dense globular star clusters—which operate on the exact same gravitational mechanics—a phenomenon called **"core collapse and halo ejection"** occurs. To reach a stable, zero-energy balance where the system doesn't immediately collapse back in on itself, it must shed the majority of its mass. The Big Pool Bangs theory applies this exact mechanism to the birth of the universe, naturally sorting cosmic mass into two distinct buckets:

- **The Ejected Halo (85%):** Roughly **85%** of the billiard balls are slingshotted outward so fast that they exceed the system's escape velocity. Because they are now flying through expanding space far apart from each other, they rarely collide again. They become an invisible, free-floating background of mass. **This perfectly explains Cold Dark Matter.** It does not require exotic, undiscovered particles; it is simply the 85% of the primordial billiard balls that got kicked off the table during the cosmic break.

- **The Trapped Furnace (15%):** The remaining **15%** of the balls did not get enough speed to escape. They fall back into the center, violently grinding and smashing against each other in a trapped, turbulent traffic jam. This violent friction generates immense heat, pulverizing the balls into the hot, glowing photon-baryon plasma that eventually forms our stars, planets, and the Cosmic Microwave Background. This represents our regular, visible universe. **Caveat (essential):** this identification requires an unspecified conversion channel from Planck relics to Standard-Model plasma — including baryogenesis and the observed photon-to-baryon ratio — while the ejected relics must remain absolutely stable for 13.8 Gyr. Reconciling relic stability (Sec. 2's incompressibility ansatz) with furnace disintegration, e.g. via a threshold collision energy that ejecta never reach, is an open requirement (Sec. 10); absent it, the 15% trapped fraction cannot be directly identified with the baryon budget.

4. Resolving Cosmological Puzzles & Deriving the Red Tilt

This mechanical rebound resolves the classical Big Bang paradoxes without requiring inflation:

- **The Horizon Problem — not solved (corrected):** an earlier draft claimed $R_{\text{jam}} \ll ct_P$; in fact $ct_P \approx 1.6 \times 10^{-35}$ m, so the jammed core is $\sim 10^{20}$ Planck lengths across and *cannot* equilibrate within a few Planck times. A bounce completing on a Planck timescale would require acausal coordination across the core; at present, homogeneity of the initial patch is an assumed initial condition, and the horizon problem remains open (Sec. 10).

- **The Flatness Problem — consistency, not yet derivation:** a *monolithic* zero-energy kick ($f = 1$ for every particle) ejects 100% and is incompatible with the 85/15 split. But the turbulent regime favored by our simulations is not monolithic: at $\bar{f} = 0.7$, $\sigma_{\text{rel}} = 1.0$ the mean kinetic energy is $\langle v^2 \rangle = \bar{f}^2(1 + \sigma_{\text{rel}}^2)v_{\text{esc}}^2 \approx 0.98\, v_{\text{esc}}^2$ — total energy $K + U \approx 0$ *while* ejecting $\sim$89%: the positive energy of the ejecta is balanced by the binding energy of the trapped furnace. The zero-energy (flat) state is thus *consistent* with the observed split in exactly the parameter regime the simulations favor; what remains missing is a derivation of *why* the rebound selects $E \approx 0$, so flatness is consistent but not yet explained.

A Phenomenological Parametrization of the Red Tilt ($n_s \approx 0.965$)

In the Big Pool Bangs theory, the observed CMB red tilt is derived directly from the dissipation rate of the **Kolmogorov energy cascade** in the turbulent quantum fluid. Energy cascades from macro-scales to micro-scales ($E(k) \propto k^{-5/3}$). During the elastic bounce, modes freeze out as they exit the acoustic horizon. The dimensionless scalar curvature power spectrum $\mathcal{P}_{\mathcal{R}}(k)$ is modified by the logarithmic decay of vortex reconnection efficiency, shifting the scalar index by ν_t (the ratio of vortex eddy turnover time to spatial expansion time):

$$n_s - 1 = -\frac{1}{3}\nu_t$$

Calculated for hard-sphere close packing ($\eta \approx 0.64$), the turbulent ratio is $\nu_t \approx 0.105$:

$$n_s - 1 = -\frac{1}{3}(0.105) = -0.035 \implies n_s = 0.965$$

This is **consistent with** the observed Planck 2018 value ($n_s = 0.965 \pm 0.004$) without requiring a scalar inflaton field. We emphasize that the relation $n_s - 1 = -\nu_t/3$ and the value $\nu_t \approx 0.105$ are at present phenomenological inputs, not first-principles results; deriving ν_t from vortex-reconnection dynamics at jamming — and showing that the Kolmogorov cascade ($E(k) \propto k^{-5/3}$) can produce a *nearly scale-invariant* curvature spectrum rather than a steeply red one — is a central open problem for the theory (Section 10).

5. Unification: Dark Energy, H_0, and Acceleration

With Dark Matter accounted for via the ejected halo, the framework also unifies the remaining cosmic phenomena:

- **Resolving the Hubble Tension (H_0):** The H_0 tension arises because space expands linearly ($a(t) \propto t$) driven by the initial kinetic rebound. Early-universe acoustic horizon scales measure global Hubble expansion, while local distance ladders measure regional kinetic drift accelerated by nearby mass sub-clusters.

- **Dark Energy & Apparent Acceleration:** Cosmic acceleration ($q < 0$) does not require an elusive cosmological constant (Λ). In a granular expansion driven by a kinetic bounce, outer galactic shells outpace inner shells due to initial velocity dispersion. As light travels across billions of light-years, Type Ia supernovae in distant galaxies appear fainter than predicted by decelerating models because space-time expansion is purely kinetic, mimicking the observational effects of dark energy.

5.1 A Natural Prediction: Dark Energy Must Evolve

The framework makes one qualitative prediction that distinguishes it sharply from ΛCDM: **the apparent dark-energy component cannot be constant.** In ΛCDM, Λ is a fixed vacuum term with equation of state $w = -1$ at all times. Here, by contrast, the apparent acceleration is a *kinematic artifact* of the ejecta velocity distribution: the effective equation of state inferred by an observer fitting FLRW models to a granular kinetic expansion is set by the evolving ratio of velocity-dispersion energy to clustered rest-mass energy. Both ingredients change with time — the dispersion redshifts and the mass distribution clusters — so the inferred $w(z)$ *must* drift. A cosmological constant is the one behavior this mechanism cannot produce.

This prediction has become newly relevant: combined DES-SN5YR (Dovekie recalibration) and DESI DR2 BAO analyses now reject a pure cosmological constant at 3.2σ in a $w_0 w_a$CDM fit, preferring $w_0 = -0.803 \pm 0.054$, $w_a = -0.72 \pm 0.21$ — i.e. dark energy that was weaker in the past and is evolving. However, the *sign* of the observed drift currently disfavors the simplest version of our mechanism: the $w_0 w_a$ best fit has w rising from ≈ -1.5 in the past toward -0.8 today (less negative with time), whereas the toy shell calculation of Sec. 9.1 drifts the opposite way (slowly more negative with time). The framework thus shares with the data the qualitative claim *dark energy evolves*, but its present toy realization predicts the wrong drift direction — by our own falsifiability criterion this counts against Hypothesis 1 unless a refined treatment (relativistic shells on an expanding background, furnace backreaction) reverses it. We stress the limits of this claim: the present framework predicts the *direction* of the drift (evolving, weaker in the past), not yet the values of (w_0, w_a); the quantitative attempts in Sec. 9.1 (Newtonian toy, relativistic LTB lightcone, profile inversion) find that ejecta streaming produces no apparent acceleration at all — so this subsection's claim survives only conditionally, on the unresolved blast-wave branch discussed there — and a mismatch in the drift's sign or magnitude would count against the model's core claims (part of Hypothesis 1, “The Big Pool Bang” — the core claim in the two-hypothesis falsifiability structure defined in Section 9, where Hypothesis 2, “The Cold Eye of Sauron — the Cold Spot and the Axis of Evil”, collects the separable bonus conjectures).

6. The ~5° Cold Spot & The Jovian/Cyclonic Vortex Model

Because vacuum crystallization occurs stochastically across an infinite background, our universe is not unique. A neighboring universe nucleating 10–12 billion light-years away creates an expanding domain boundary wall. Simultaneously, internal quantum turbulence in our own bounce creates macro-vortices.

Rather than treating CMB anomalies as abstract spherical collisions or Gaussian flukes, the **Jovian/Cyclonic Model** maps early-universe quantum fluid dynamics directly to planetary atmospheric mechanics.

1. **The Vortex Core Choke:** Macro-scale quantum vortex tangles act as thermal dams, choking off local kinetic heat transport in the plasma.

2. **Asymmetric Blob Formation:** Due to fluid shear and vortex stretching, these primordial cyclonic storms deform into elongated “blobs” aligned along a preferred rotation axis (the “Axis of Evil”).

3. **Frictionless Persistence:** Without solid-surface friction to dissipate angular momentum, these macro-vortices scale up with spatial expansion, freezing into the CMB sky at decoupling as features like the **~5° radius Eridanus Cold Spot** ($\Delta T \approx -70\,\mu$K).

Key Prediction: If the Cold Spot is a primordial cyclone, thermal energy diverted around its core will form a concentric hot ring, accompanied by a sharp radial **E-mode polarization ring** along its boundary—a primary test for upcoming CMB arrays.

(Positional note: the Cold Spot is static for all practical purposes — any intrinsic transverse motion of its source, under either the cyclonic or cross-boundary reading, corresponds to nanoarcseconds per century of angular drift, so proper motion cannot discriminate Hypotheses 2 and 3. The only measurable drift of any CMB feature is apparent: the changing aberration from our own acceleration, of order microarcseconds per year, whose detection — already demonstrated for quasars with Gaia astrometry — is a goal of real-time cosmology but is a statement about us, not the Spot.)

6.1 Independent Evidence for a Preferred Axis (external literature)

A preferred axis could arise in this framework only from a seed asymmetry of the initial patch (e.g. net angular momentum) — our own simulations find ejection isotropic within shot noise (Sec. 9.1), so the framework currently provides *no* mechanism for a preferred axis, and this subsection is motivation for seeking one rather than support. Suggestively, the existing literature — independent of this work — reports a cluster of tentative alignments in one region of the sky: (i) a dipole anisotropy in the cosmic acceleration inferred from SNe Ia, directed roughly along our motion relative to the CMB [Colin et al. 2019; Mariano & Perivolaropoulos; see also Sah & Rameez 2024, arXiv:2411.14758; Union2 hemisphere analyses, MNRAS 446, 2952]; (ii) the alignment of the CMB low multipoles — the quadrupole/octupole “Axis of Evil” — with the dipole direction, reported at $> 99\%$ confidence in moving-dark-energy analyses [arXiv:0811.3606]; (iii) bulk-velocity-flow directions and the quasar optical-polarization alignment axis falling in the same angular region, a coexistence judged statistically unlikely to be coincidental [arXiv:1401.5044]. Each signal is individually sub-decisive ($\lesssim 3\sigma$) and contested — peculiar-velocity corrections absorb part of the SN dipole — but their spatial coincidence is exactly the morphology expected if one physical cause, such as uneven primordial ejection, sets a common axis for the expansion anisotropy, the bulk flow, and the large-angle CMB pattern. We cite these alignments as motivation for Hypothesis 2, "The Cold Eye of Sauron — the Cold Spot and the Axis of Evil" (the separable bonus conjectures defined in Section 9), not as confirmation. One adjacent anomaly class does *not* join the cluster: claimed galaxy spin-handedness dipoles (Longo 2011 at galactic (52ř, +68.5ř); Shamir 2012–2022 at (192ř, +38ř)) lie 70ř from *each other* and ≈50ř–74ř from the Axis-of-Evil and flow axes — mutual inconsistency that supports the systematics reading (Iye et al. 2021) rather than a cosmological spin axis; we checked and report the null.

7. Black Hole Collision Signatures (Universe 1 vs. Universe 2) — Hypothesis 3, "Big Pool Bang Multiverse 2 in the Cold Spot?"

If black holes from Universe 2 cross the domain boundary into our expansion sector, their physical properties differ sharply from our local primordial black holes:

Physical Feature	Universe 1 Black Holes (Local Primordial)	Universe 2 Black Holes (Cross-Boundary Infall)
Mass Profile	Capped at Planck relics (m_P) or power-law clusters.	Standard stellar/supermassive spectrum, with sub-solar anomaly gaps ($0.1 - 1.0 M_\odot$).
Sky Distribution	Isotropic (evenly distributed across dark matter halos).	**Anisotropic**; heavily clustered toward the ~5° Cold Spot direction.
Peculiar Velocity	Bound to local galactic orbital speeds ($v \sim 200$ km/s).	**Velocity Kick:** High peculiar drift ($v > 1000$ km/s) directed from the Cold Spot.
GW Waveform	Standard orbital inspiral "chirp".	**Gravitational Wave Memory:** Step-function metric strain (Δh).

7.1 The Spin Constraint on Universe 2

A neighbor universe is the only agent that can torque ours (internal angular momentum cancels by conservation), so the measured non-rotation of the universe constrains Hypothesis 3 quantitatively. Order of magnitude: an equal-mass neighbor ($M \sim 1.5 \times 10^{53}$ kg) at the Cold Spot distance ($D \sim 10$–12 Gly) exerts a tidal field $T \sim GM/D^3 \approx 10^{-35}\,\mathrm{s}^{-2}$ — of order H_0^2, i.e. raw coupling of order unity. Acting on our mass distribution's large-scale quadrupole asymmetry ($q \sim \delta \sim 10^{-5}$) over the causally available few Gyr ($t_{\mathrm{int}}/t_H \sim 0.2$), the induced vorticity is $\omega/H_0 \sim (T/H_0^2)\, q\, (t_{\mathrm{int}}/t_H) \approx 2 \times 10^{-6}$ — **three orders of magnitude above the Planck Bianchi bound** $\omega/H_0 \lesssim 10^{-9}$. A Cold Spot neighbor of comparable mass therefore requires at least one of: a domain boundary transmitting $\lesssim 0.1\%$ of tidal shear (a calculable property of the vacuum wall, presently uncomputed), a several-fold larger distance (the D^{-3} scaling recovers two orders at $\sim$40 Gly), or a much smaller neighbor mass. The same tidal field also imprints a direct CMB quadrupole through which the identical constraint applies a second time. **Mass estimate from the Cold Spot itself.** If the Spot's decrement is the neighbor's potential imprint, Sachs–Wolfe inverts it into a mass: $\Delta T/T \approx 2.6 \times 10^{-5} \Rightarrow \Phi/c^2 \approx$

8×10^{-5}, giving $M_2 \approx 3(\Delta T/T)\, c^2 D/G \approx 10^{49}$ kg at $D \approx 10$ Gly — a *dwarf* neighbor, $\sim 10^{-4}$ of our universe's mass (angular-size check: 5ř at 10 Gly subtends $\sim$0.9 Gly, a plausible imprint scale). Fed back into the torque estimate, which scales linearly with mass, this gives $\omega/H_0 \approx 1.4 \times 10^{-10}$ — *under* the Planck bound with a factor $\sim$7 margin. The mass the Cold Spot's amplitude demands is automatically small enough to satisfy the non-rotation constraint: Hypothesis 3's first two-observable consistency pass, subject to $\mathcal{O}(1)$ Sachs–Wolfe factors, the assumed D, and the uncomputed boundary transmission (which suppresses the imprint, pushing M_2 up and eroding the margin). The global spin bound remains the strongest quantitative constraint on Hypothesis 3 — independently enforcing the distant/weakly-coupled regime that the Planck spot-count statistics already impose on Hypothesis 4 (P7), and consistent with the torque-axis near-null of Sec. 6.1 (the Axis of Evil lies 56ř, not the torque-predicted 90ř, from the Cold Spot).

The spot population as a bubble spectrometer. The two observables of any anomalous spot invert into the neighbor's parameters: Sachs–Wolfe gives $M/D \propto \Delta T$, and an imprint scale set by the neighbor's own size ($R \propto M^{1/3}$ at fixed formation density) gives $\theta \propto M^{1/3}/D$ — hence per spot

$$M \propto (\Delta T/\theta)^{3/2}, \qquad D \propto \Delta T^{1/2}\, \theta^{-3/2}.$$

This applies only to the *non-Gaussian outlier* population (the acoustic-peak spots are ΛCDM's best-confirmed physics and are not available for reassignment), so current statistics are poor ($n \sim 3$–5 candidate features, with look-elsewhere hazards). But the population-level prediction is sharp: neighbor bubbles drawn from a first-order nucleation spectrum populate a correlated power-law *locus* in the $(\Delta T, \theta)$ plane, whereas Gaussian random-field peaks obey BBKS statistics with amplitude and size nearly independent — a computable discriminant on existing Planck maps. Most importantly, the inferred neighbor mass function *is* the nucleation size spectrum of the vacuum transition — the same distribution that, inside our own bang, drove the cascade of Sec. 1 and set the kick statistics behind the dark-matter fraction (Sec. 9.1). One mechanical process, sampled internally (ejecta kinematics) and externally (spot statistics): if both trace one transition, the two inferred bubble-size distributions must agree — a cross-consistency requirement linking Hypotheses 1 and 4 that either dataset can falsify.

Confrontation with data (locus test, this work). We ran the discriminant on the Planck SMICA map ($N_{\rm side} = 128$, $|b| > 25$ř, 2ř smoothing): the eight most extreme large-scale spots were extracted with per-spot amplitude ΔT and half-maximum radius θ, and the amplitude–size scaling was fit in log–log space; sixty Gaussian realizations drawn from the map's own pseudo-C_ℓ were passed through the identical pipeline as the null. The extreme spots (which include the Cold Spot core at $(l, b) = (203$ř$, -56$ř$)$, $\Delta T = -168\,\mu$K, $\theta = 2.5$ř) span $\Delta T = 153$–$170\,\mu$K and $\theta = 2.5$ř–5.5ř, and yield a slope $d \ln |\Delta T|/d \ln \theta = -0.00$ (correlation $r = -0.01$), sitting at the 27th percentile of the Gaussian null ($+0.07 \pm 0.10$). No amplitude–size correlation of any sign is detected, consistent with Gaussian peak statistics — but an adversarial re-examination of the pipeline (this work) found the test has little power as implemented: the 2ř smoothing quantizes the recoverable radii to the 2.5ř–5.5ř range and the top-8 selection compresses the amplitude spread to $\sim 10\%$, so a near-zero slope is the likely outcome under *both* hypotheses. We therefore report the locus test as *unresolved pending injection-recovery validation* (simulate a $\Delta T \propto \theta^3$ population, push it through the pipeline, verify the slope is recoverable), not as an exclusion. This does not falsify H4 outright — neighbors at a *range* of distances with a compensating mass spectrum can populate a diffuse cloud rather than a tight locus, and a single genuine neighbor among seven Gaussian impostors would not move the fit — but it removes any positive population-level evidence: nothing in the current extreme-spot sample requires a second bang beyond the Cold Spot's own (independently argued) anomaly. One observation, however, survives (with caveats): *all eight* extreme spots lie in the southern galactic hemisphere. In

200 Gaussian isotropic realizations passed through the same pipeline, not one placed its full top-8 in a single hemisphere (mock mean: 5.2 of 8); the run breaks at rank 9, and the top 15 remain 12-of-15 southern. Honest bounds on this: 0/200 licenses only $p < 1/201$, our mocks are simplified (unmasked pseudo-C_ℓ, no anisotropic noise or foreground residuals, hemisphere frame chosen post hoc), and if the Cold Spot and its two nearby hot extremes are one physical system (see below) the count is effectively 6/6, weakening the statistic. The published assessments of the hemispherical power asymmetry ($p \sim 1$–10%, estimator-dependent; Planck 2015/2018 isotropy papers) are the appropriate significance to carry. This is the well-known hemispherical power asymmetry rediscovered through the extreme-spot population — but in the present framework it is not a separate anomaly: a single massive neighbor to the (galactic) south, the same object invoked for the Cold Spot in H3, modulates large-scale power on one side of the sky by tidal and Sachs–Wolfe action, so the asymmetry, the Cold Spot, and the preferred axis of Sec. 6.1 become one prediction with one direction, rather than three coincidences. An isotropic swarm of neighbors (H4) predicts no such one-sidedness.

The vortex reading of the border shell. An alternative to the neighbor-window interpretation treats the extreme spots as *relic vortices* of the bang's own fluid phase, riding the earliest, fastest ejecta to the border. A vortex imprints temperature twice over: its core is a pressure minimum — rarefied, adiabatically cooled, ringed by compressed warmer material (the Cold Spot's anomalous cold-core/hot-ring template is exactly this morphology) — and its rotation adds a Doppler *dipole* across the spot whose orientation encodes the projected spin axis. Because angular momentum is conserved while rotational velocity decays ($v_{\rm rot} \propto 1/a$ against $r \propto a$), the spin never vanishes; and because homologous expansion preserves angular geometry, the border shell — which is what the CMB sky *is*, seen from near the center — should carry the fossil arrangement of the original vortex configuration. Notably, in this reading no *nearby* mass counterpart is expected (the slow, late, spin-poor ejecta stayed near the center where we sit), which is consistent with our 2M++ cross-match of the eight sightlines: nearby ($< 180\,h^{-1}$Mpc) structure matches the spot signs at coin-flip rates (5 of 8, with two cold spots behind strong *over*densities), and the only confirmed deep counterpart anywhere on the sky remains the Eridanus supervoid at the Cold Spot — itself too shallow to explain the decrement.

Confrontation with data (this work). We measured each extreme spot's internal dipole (monopole+dipole fit within 6ř, local tangent frame) and ran the vortex-configuration tests against the same 100–200 Gaussian nulls. Results: (i) *hot/cold balance* — the eight extremes split 4/4 with rank-matched amplitudes summing to $-11\,\mu$K against a mean $|\Delta T| \approx 162\,\mu$K; only 7% of Gaussian skies are as balanced — suggestive of cancelling pairs, short of a detection. (ii) *Spatial pairing* — opposite-sign spots lie 4.0ř closer than same-sign (null +5.2ř ± 16.3ř): the vortex-dipole sign, at $p = 0.30$. (iii) *No lattice*: nearest-neighbor spacing is no more regular than Gaussian. (iv) *Spin budget* — converting dipoles to spin-axis proxies ($\mathbf{L} \propto \hat{c} \times \mathbf{d}$), the Cold Spot's internal dipole ($16.4\,\mu$K/deg) is the largest in the sample by a factor 2.3, and the remaining spots' proxies weakly co-align rather than counter-rotate. However, a monopole+dipole fit over a 6ř disc on a 2ř-smoothed map is dominated by $\ell \sim 2$–20 gradients that Gaussian peaks generically carry, correlated across co-hemispheric spots by shared low-ℓ modes — so we do **not** advance these numbers as spin measurements; they are upper bounds on any rotational contribution, and the co-alignment is most simply gradient leakage. A gradient-subtracted re-fit (removing the best-fit $\ell \leq 10$ sky first) is the required upgrade before any spin fraction can be quoted. The clean channel is curl: paired relic vortices predict zero *mean* but excess curl-mode *variance* (lensing curl, B-mode polarization) localized at the extreme spots, where ΛCDM predicts the curl null holds everywhere — a decisive test on existing Planck lensing products. (v) *Configuration*: two of the four hot extremes lie 11ř–12ř from the Cold Spot, near (though inside) its literature hot-ring radius of $\sim$ 15ř; they *may* be pieces of one system — but we flag the tension this creates: if merged, every $n = 8$ statistic above must be recomputed with $n = 6$, and no null was computed for the annulus coincidence itself, so we carry the fragment reading as a hypothesis, not a result. *Statistical accounting.* Across the full battery reported in this subsection (locus slope, hemisphere count, hot/cold balance, pairing distance, lattice regularity, dipole magnitudes, cancellation ratio, κ stack — nine statistics), the individual results are $p \sim 0.06$–0.7; under any global look-elsewhere correction the sub-2σ items are jointly consistent with noise, and we say so plainly. What survives is qualitative: the extreme-spot sky is one-sided at literature-level significance, the Cold Spot is the outlier in every channel examined, and nothing requires a population of additional objects. The **load-bearing, falsifiable claim of the vortex reading is the curl test (P9)**: we commit in advance that a clean stacked curl-mode null at the Cold Spot position, at sensitivity reaching the temperature-dipole-implied rotation, *retires* the vortex interpretation of that direction in favor of the non-rotating neighbor-shadow reading (H3) — not merely disfavors it. If instead curl variance is detected localized there, the vortex reading stands and the seed-side interpretation of $(l, b) \approx$ (200ř, −50ř) becomes the economical account.

Why only the top layers carry the fossil. The velocity sorting of Sec. 9.1 is statistical: only the earliest, fastest ejecta escaped in a single pass, preserving delicate geometric correlations; everything slower underwent violent relaxation — multiple core crossings that conserve spin *magnitude* but randomize spin *geometry.* The model therefore predicts a **coherence gradient**: fossil vortex geometry intact at the border shell, degrading rapidly inward, absent in the well-mixed interior — with inner layers retaining at most statistical residues (a slight net handedness with no preferred direction). Two archival tests agree with this. (i) Intermediate-shell spin claims do *not* point at the spot cluster: the quasar-polarization alignment regions of Hutsemékers et al. ($z \sim 1$–2, coherent over ∼Gpc, with the alignment angle twisting ∼ 30ř/Gpc in depth — phenomenologically a differentially rotating medium, though contested) and the Longo/Shamir galaxy-handedness axes lie 24ř–73ř from the (200ř, −50ř) direction, consistent with random axes — an expected null under mixing, since these tracers inhabit the relaxed layers. (ii) We stacked the public Planck PR4 lensing convergence reconstruction (whose kernel peaks at $z \sim 2$, squarely in the mixed zone) on the eight spot positions: the sign-weighted stack is $+0.006$ against a random-position null of 0.000 ± 0.013 (68th percentile; 4/8 sign matches), i.e. no integrated mass scars along the sightlines — again as mixing predicts. The Cold Spot's $\kappa = -0.050$ (-1σ) has the sign its known shallow foreground supervoid requires, but at 1σ we do not promote it: by the same standard applied to the other seven sightlines it is noise. The public lensing release contains only gradient (convergence) modes; the curl-mode stack of P9 — the decisive geometry-capable probe at the shell itself — awaits either the unreleased curl reconstructions or a dedicated re-analysis. We also ran the polarization version directly on the SMICA Q/U maps ($N_{\rm side}$ 256, $\ell \leq 512$, E/B decomposition, 1ř smoothing): the stacked B-mode variance in 6ř discs at the eight spots sits at the 66th percentile of random positions, the E-mode variance at the 35th, and the Cold Spot's radial-E (stacked Q_r) profile is consistent with zero in every 2ř ring to 20ř — but with per-ring uncertainties of 0.6–2.7 μK, the predicted 1–3 μK polarization ring (P1) is neither detected nor excluded: Planck's polarization noise at these scales cannot yet test it. P1 and P9 therefore remain open, awaiting CMB-S4/LiteBIRD sensitivity.

A first direct curl search (this work). Since the public lensing release lacks curl modes, we implemented an approximate temperature-only quadratic estimator ($N_{\rm side}$ 256, $\ell \leq 512$, inverse-variance × Wiener-gradient product, spin-1 curl projection, mean field subtracted using 40 Gaussian simulations through the identical masked pipeline; unnormalized, so valid only as a relative comparison). Stacking the curl-potential variance in 6ř discs: the eight spots sit at the 82nd percentile of 60 simulated skies, and the Cold Spot itself at the **38th percentile** — its curl content is indistinguishable from a random position. This is the first direct curl look at these positions and it finds nothing. It does *not* yet trigger the retirement clause of P9: the commitment is bound to a sensitivity reaching the temperature-dipole-implied rotation, which this reduced-resolution, uncalibrated estimator demonstrably does not attain (Planck's full-depth reconstruction uses $\ell \leq 2048$ with polarization). The vortex reading survives, weakened: the cheap version of its decisive test came back empty, and its remaining life expectancy is set by CMB-S4/LiteBIRD.

Spiral winding and the mass profile (this work). A spinning cap must wind its internal pattern: azimuthal features at successive radii should be phase-rotated by the differential rotation, saturating at the rim where the vortex velocity falls off ($v \sim 1/r$ outside a solid-body core). Measuring the azimuthal phase of the m=1 mode in 1.5ř rings from 1.5ř to 15ř around the Cold Spot: the phase is locked ring-to-ring with coherence 0.988 — **higher than all 300 random positions tested** (null 0.72 ± 0.18) — and drifts at a slow $\sim$ 3ř of phase per degree of radius ($\sim$ 40ř total twist). A static background gradient produces high m=1 coherence but *zero* winding; the Cold Spot shows both, and only the winding is spin-specific. The m=2, 3 modes are unremarkable, and this is a post-hoc test subject to the same look-elsewhere discount as the rest of the battery. In the lensing mass (PR4 κ, 2ř ring profile): a concentrated central deficit, $\kappa = -0.070$ vs null $+0.013 \pm 0.042$ (-2σ) in the inner 2ř, decaying by 4ř — the sightline's integrated mass hole, consistent with the foreground supervoid — but **no "belt"**: the 8ř–20ř annulus never exceeds $+0.6\sigma$ (the largest bump does sit at 14ř–18ř, the temperature hot-ring radius, at noise level). Physically consistent: a shear rim concentrates velocity, not necessarily mass, and condensing a mass ring requires dissipation the model does not supply.

The small-spot population and the necessity of CMB-independent mass tracers (this work). Stacking the Planck PR4 lensing convergence on the ~27,000 ordinary small spots (1ř smoothing, $|T| > 1.5\sigma$ extrema) initially showed a striking temperature–mass correlation: cold spots on mass deficits (-7.2σ), hot spots on excesses ($+2.2\sigma$), scaling with spot amplitude — the signature expected of ISW modulation (and of dark energy). But the κ map is reconstructed *from the same CMB temperature data*, and quadratic estimators leak temperature features into the mass map. Repeating the stack against a fully CMB-independent mass tracer — the Quaia Gaia–unWISE quasar catalog (756k quasars, $z \sim$ 1–3, selection-function corrected, 52% sky) — **the correlation vanishes**: cold spots -1.1σ, hot -0.3σ, no amplitude scaling; the eight extreme spots are likewise within $\sim 1\sigma$ of random quasar density (the Cold Spot mildly negative, -0.018, 0.6σ). The κ correlation is therefore *either* reconstruction leakage (the κ map is a quadratic functional of the same temperature data, biased at temperature extrema) *or* the genuine ISW×lensing correlation (an established Planck-era effect) — and an independent re-review of this chain (this work) correctly notes we have not discriminated between them: the Quaia null is not a *sensitive* null (its $z \sim$ 1–3 kernel barely overlaps the $z \lesssim 1$ ISW kernel, and its shot noise per disc exceeds the expected ISW-scale overdensity), and no fiducial-ISW forecast was computed for either tracer. The decisive archival test is a polarization-only κ stack (no temperature in the estimator), which we have not performed. One lesson stands regardless: every CMB-lensing-based "mass" statement in this section carries a *CMB-derived* asterisk, with only the 2M++ ($z \lesssim 0.07$), Quaia ($z \sim$ 1–3), and DES-literature results truly independent. We then closed that loophole with the WISE×SuperCOSMOS photometric-redshift catalog (17.7M galaxies at $z = 0.1$–0.4, server-side aggregated, high-passed against a 10ř baseline to remove depth gradients — CMB-independent). The *small*-spot stack remains null there (cold -0.7σ, hot $+0.2\sigma$) — a measurement, not yet an attribution (see above). The *extreme* spots show a suggestive pattern at ISW depths: with a correlated-noise null from 300 random 5ř discs, seven of eight match sign (cold spots on galaxy deficits, hot on excesses), combined sign-weighted stack $\approx +2.7\sigma$ — including two sightlines whose *shallow* ($< 180\,h^{-1}$Mpc) structure had the wrong sign. Mandatory qualifications from independent review: the result rests on three sightlines ($|z| \approx 2.0$–2.3; the other five average $+0.6\sigma$ combined, and four "matches" are near-coin-flips); the nominal binomial 3.5% ignores the three-tracer, multi-statistic search (global $p \sim 10$–30%); the spots were selected as temperature extremes partly *because* any foreground contribution deepened them, so extremeness-conditioning inflates the recovered correlation (Eddington-type bias); the 10ř high-pass baseline partially self-subtracts genuine $\gtrsim$ 10ř supervoids (attenuating the Cold Spot's own -0.6σ) and can imprint compensating rings; and WISE×SuperCOSMOS extinction/stellar residuals at $|b| = 25$–40ř are of order the per-spot signals. Net verdict, stated before further data: this is a $\sim$ 1–2σ-class hint that the extreme spots are partly sourced by $z \sim 0.1$–0.4 superstructures (the Eridanus story generalized), to be confirmed or retired by a pre-registered re-run at two baseline scales with $|b| > 30$ř and extinction regression — not yet grounds to weaken or support either the border-shell or the en-route reading. The Cold Spot itself is only -0.6σ here, its known too-shallow-void tension intact.

The JWST handedness coincidence. One archival dataset lands directly on the column sightline: the JWST/JADES deep field — which lies at $(l, b) = (224$ř$, -54$ř$)$, only 8.6ř from the Cold Spot — shows a claimed $\sim$2 : 1 spin-handedness asymmetry in 263 high-redshift galaxies (66% clockwise; Shamir 2025, MNRAS). This is the deepest galaxy-spin measurement in existence, and it sits, by accident of survey design, on the one direction where the vortex reading predicts residual handedness in the upper layers. We weight it cautiously: the sample is small, the classification method is contested (independent reanalyses of analogous ground-based claims found no asymmetry), and a mundane candidate exists (a Doppler boost from the Milky Way's own rotation, maximal near the galactic poles). The discriminating measurement is a *control field*: the identical pipeline on JWST's COSMOS or GOODS-N fields ($>$ 100ř away) — a systematic reproduces the asymmetry everywhere; a Cold Spot spin column produces it only on the column. That comparison is not yet published and is, alongside the curl stack, the second decisive archival-era test of this subsection. The core-side anchor of the coherence gradient is already measured (this work): the same-handedness angular correlation function over 77,840 SDSS spin-annotated galaxies (11.2M pairs, 0.05ř–4ř) is zero in every bin to a few parts in 10^3, and the global handedness excess is 1.9σ (50.34% clockwise) — the local, well-mixed layers carry no direction memory, exactly as mixing requires, while the claimed coherence lives only far up the column.

Method caveat (adversarial review, this work). An internal adversarial review of this subsection's pipeline identified, beyond the items already folded in above: the Gaussian nulls use the unmasked full-sky pseudo-C_ℓ without mask deconvolution, anisotropic noise, or foreground residuals (end-to-end NPIPE/FFP simulations are the required upgrade for any sub-percent claim); the spots are selected on the same map on which all subsequent statistics are evaluated, so the hypotheses tested were partly formulated post hoc (a clean version would formulate on one component-separation map and test on another); and the "coherence gradient" argument was introduced after the intermediate-shell nulls it explains, which is why we bind the vortex reading's falsifiability to the pre-registered curl commitment above rather than to any of the reframed nulls.

8. Experimental Noise Levels & Measurement Thresholds

Detecting cross-boundary black hole collisions requires isolating a non-oscillating gravitational memory strain ($\Delta h \sim 10^{-22} - 10^{-21}$) from noise. Quantum shot noise and the cosmic binary merger background present significant hurdles. Current ground interferometers (LIGO/Virgo/KAGRA) lack the low-frequency response ($<$ 10 Hz) and sensitivity needed to separate step-function memory strain (Δh) from environmental seismic noise, necessitating specialized facilities.

9. Falsifiable Predictions and Required Instrumentation

The theory's claims fall into four hypotheses. **Hypothesis 1 — "The Big Pool Bang" (core, must hold):** the dark-matter and dark-energy phenomenology — the ejected-halo dark-matter fraction (P6, P3) and the kinetic expansion history reproducing the observed acceleration and Hubble-constant behavior (P5, Sec. 9.1). **Hypothesis 2 — "The Cold Eye of Sauron — the Cold Spot and the Axis of Evil" (bonus, independent of the core):** the Cold Spot cyclonic-vortex interpretation and the Axis-of-Evil alignment (P1, P2). **Hypothesis 3 — "Big Pool Bang Multiverse 2 in the Cold Spot?" (bonus, independent of both):** a neighboring Big Pool Bang universe (Universe 2) nucleated beyond the domain boundary in the Cold Spot direction, signaled by cross-boundary black-hole infall, the directed dark-flow component, and gravitational-wave memory (P4; Sections 6–8). **Hypothesis 4 — "More Multiverses Hiding in the Dark Comic Background?" (bonus, generalizes Hypothesis 3):** since vacuum crystallization is stochastic across an unbounded background, our neighbor should not be unique — a *population* of Big Pool Bang universes at varying distances, each imprinting a boundary signature as a distinct CMB anomaly (P7). Hypotheses 2–4 are separable conjectures — any can fail without falsifying the core bounce-and-ejection mechanism, and each can hold without the others (a vortex needs no neighbor; a neighbor needs no vortex; and Hypothesis 4 stands or falls on population statistics, not on any single spot). For each prediction we state the model's value (order-of-magnitude at this stage), the sensitivity required, and the instrument that reaches it:

#	Prediction	Observable & required threshold	Instrument
P1	Concentric hot ring around the Cold Spot with amplitude $\Delta T \approx +20$–$50\,\mu$K (energy diverted from the $-70\,\mu$K core), plus a co-located radial E-mode ring of ~ 1–$3\,\mu$K. **Tested against Planck 2018 data in this work (Sec. 9.1): no ring at the naively estimated** 5ř–7ř**; a $+37$–$52\,\mu$K excess appears at** 13ř–15ř **but only at** $\sim 2.5\sigma$ **and may be a fluke. The thermal ring is an *optional* signature — a Jovian “eye of the storm” (calm cold core, no hot ring) is equally consistent with the cyclonic picture. The decisive signature is the polarization structure (below)**	Polarization sensitivity $\lesssim 1\,\mu\mathrm{K}\cdot$arcmin at $\ell \sim 20$–100, over a $\gtrsim$ 30ř patch	CMB-S4
P2	Cyclonic B-mode turbulence pattern aligned with the “Axis of Evil”	B-mode maps at degree scales, r-equivalent sensitivity $\sim 10^{-3}$	CMB-S4, LiteBIRD
P3	Sub-solar black hole mergers (0.1–1.0 $M_\odot$), anisotropically clustered toward the Cold Spot	Strain $\sim 10^{-24}/\sqrt{\mathrm{Hz}}$ at 10–10^3 Hz; sky localization $\lesssim 10\,\mathrm{deg}^2$	LIGO/Virgo/KAGRA O5, Einstein Telescope, Cosmic Explorer
P4	Gravitational-wave memory (step-function strain $\Delta h \sim 10^{-22}$–10^{-21}) from cross-boundary infall	Sub-Hz response with Δh sensitivity $\lesssim 10^{-22}$; free of seismic wall < 10 Hz	Einstein Telescope (underground), LISA (mHz)

#	Prediction	Observable & required threshold	Instrument
P5	Coherent bulk flow of $\sim$400 km/s persisting (not converging) at bias-corrected depths $\gtrsim 200\,h^{-1}$ Mpc along the flow axis ($l \approx$ 298ř, Watkins et al. 2023) — amplitudes $\gtrsim$ 600 km/s at these depths are already disfavored by the Planck kSZ limit	Peculiar-velocity survey to $z \sim 1$ with per-cluster errors $\lesssim$ 300 km/s	Rubin/LSST + Euclid cross-correlation

Locating Ourselves: the Heading and Distance of the Bang's Center

Unlike standard cosmology, this framework has a center (the seed bubble / furnace), so "where is it?" is a meaningful question. We stress at the outset that the *machinery* of this section is not new: Mpc-scale off-center observer constraints from the CMB dipole are standard in the void-model literature (Alnes & Amarzguioui 2006, $\lesssim$ 15 Mpc; Foreman et al. 2010; Grande & Perivolaropoulos 2011), the interpretation of the dipole anomalies as one peculiar-velocity vector is the tilted-universe program (King & Ellis 1973; Tsagas; Asvesta et al. 2022), a combined preferred axis was published by Antoniou & Perivolaropoulos (2010: mean axis (277ř, +44ř), chance probability $\sim$0.8%), and shell-wise bulk-flow fits are standard since Watkins, Feldman & Hudson (2009). What is specific to this framework is only the *attribution*: the shared vector as our displacement from a physical origin point.

The estimate. Displacement converts from velocity via the local gradient ($\approx H_0$): attributing the full CMB dipole (370 km/s) to off-centering gives $\Delta r \approx 5$ Mpc; the residual unexplained by local gravity gives $\Delta r \lesssim$ 1–2 Mpc. (A 2026 DESI DR1 tomographic analysis independently confirms the dipole's kinematic character, consistent with this cap.) The direction-dependent Hubble constant follows, $\Delta H/H \approx v/(H_0 d)$, with the discriminant in its distance behavior: kinematic dipoles decay as $1/d$; a cosmological off-centering gradient persists.

Axis audit — corrected after adversarial literature review. An earlier draft counted four "independent" anomaly axes agreeing at $p \sim 3 \times 10^{-3}$; that statistic does not survive scrutiny and is retracted. The Kashlinsky dark-flow measurement was not confirmed by the Planck team's reanalysis (Planck Int. XIII 2014: bulk flow consistent with zero, 95% limit $\sim$254 km/s — well below the claimed 600–1000); the Colin et al. q-dipole is CMB-dipole-aligned essentially by construction and is rebutted by Rubin & Heitlauf (2020), with newer Pantheon+ isotropy analyses finding no significant dipole after corrections; and the Migkas H_0 anisotropy is, by Migkas et al.'s own analysis, degenerate with a $\sim$900 km/s bulk flow — under this framework's own reading it is the *same* velocity field as the direct flow measurements, so counting both is double-counting (FLAMINGO simulations further show ΛCDM flows generically induce $\sim$3% apparent H_0 anisotropy). What survives is **two** quasi-independent data: the CMB dipole (264ř, +48ř) and the bias-corrected Cosmicflows-4 bulk flow (Watkins et al. 2023: 419 ± 36 km/s at $200\,h^{-1}$ Mpc toward (298ř ± 5ř, −8ř ± 4ř), a rising amplitude in $\sim 4.8\sigma$ nominal tension with ΛCDM, though Whitford et al. 2023 argue the estimator uncertainties are understated). These two axes lie $\sim$40ř–50ř apart — a *misalignment* the pure-offset reading must confront, since a single offset vector predicts exact coincidence; local-attractor infall (Shapley/Great Attractor; cf. "Origins of the Bulk Flow," 2025) explains the flow direction without any offset at all. The genuinely independent adjacent anomaly is the quasar number-count dipole (Secrest et al. 2021/2022, 4.4–4.9σ excess amplitude near the CMB dipole direction; Dam et al. 2023 concur; Abghari et al. 2024 and later systematics studies substantially weaken it) — an amplitude anomaly about the kinematic interpretation, cited here as contested support, not proof.

Crust fit (this work; uncorrected — illustrative only). Our own radial-shell fit to the CF4 group catalogue (38{,}050 groups) finds crust dipoles at (295ř, +16ř), (260ř, +26ř), (281ř, +26ř), (290ř, +31ř) for shells spanning 5–180 Mpc with amplitude growing 315 $\to$ 507 km/s, and a joint direction (287ř, +18ř), $\Delta r \approx$ 4–5 Mpc. This fit applies **no inhomogeneous-Malmquist correction in any shell** (beyond $\sim$180 Mpc it destabilizes entirely), and its latitude sits $\sim$26ř north of the published bias-corrected direction; the Watkins et al. numbers above, not ours, should be taken as the reference. The qualitative agreement (one axis across crusts, growing amplitude) survives; the specific coordinates do not carry independent weight.

What would decide it. The framework's residual, testable content here is: (i) the deep flow should stay coherent (not converge) at bias-corrected depths — current data lean this way but with disputed error bars; (ii) the CMB-dipole-vs-flow misalignment must be explained (e.g. a local-gravity component atop the offset) or it falsifies the single-vector reading; and (iii) the dipole-locked redshift *quadrupole* (whose sign also fixes which end of the axis holds the center) — noting Tsagas' tilted-universe work also develops quadrupole signatures, so priority requires a comparison. Within the framework, the tentative reading — with all caveats above — remains a displacement of a few Mpc along an axis in the $l \approx$ 264ř–298ř quadrant; the earlier draft's confident "return address" is downgraded to a hypothesis awaiting the bias-corrected and quadrupole tests.

Local flow-center fit (this work). A displaced expansion center is invisible in a purely linear flow (every comover sees a pure Hubble law) but detectable through nonlinearity, which enters at $\mathcal{O}(Hc^2/r)$ — $\sim$90 km/s at 10 Mpc for a 5 Mpc offset, above the famously cold $\sim$50 km/s local dispersion. Fitting the 2{,}112 CF4 groups at 1.5–40 Mpc with a radial flow $v(R) = hR + kR^2$ diverging from a free center $\vec{c}$: the displaced model beats the us-centered null by $\Delta\chi^2 = 6595$ for 3 extra parameters, with best fit $|\vec{c}| = 6.1$ Mpc toward $(l, b) =$ (118ř, −24ř) — 12ř from the *antipode* of the large-scale flow axis. Read within the framework: the local field diverges from a point on the same axis as every other tracer, on the far side — we are moving *away* from the center, resolving the sign ambiguity in favor of the velocity-sorted outward gradient (and $k < 0$, a locally decreasing expansion rate, is consistent with sorted shells). Read skeptically: the fit is uncorrected for known local dynamics — Virgocentric infall (the fitted center is 130ř from Virgo, so not simply that) and above all the documented $\sim$250 km/s push away from the Local Void, which mimics divergence-from-a-point by construction; the enormous $\Delta\chi^2$ surely absorbs real structure. The decisive follow-up is to subtract a standard local-velocity-field model (Virgo + Local Void + Great Attractor) and test whether a residual displaced-center signal survives — noting the framework-internal possibility that the "Local Void" and the bang's evacuated center are not independent explanations but the same object under two descriptions.

Concentric-shell control. To test the Local Void explanation, the same free-center fit was repeated in independent crusts. The three well-measured shells return a *consistent* center: 1.5–40 Mpc gives 4.9 Mpc toward (122ř, −20ř) ($\Delta\chi^2 = 1277$); 40–80 Mpc gives 2.4 Mpc toward (110ř, −8ř) ($\Delta\chi^2 = 73$); 80–150 Mpc gives 5.9 Mpc toward (109ř, −14ř) ($\Delta\chi^2 = 229$) — directions agreeing within $\sim$10ř–15ř across shells, two of which lie entirely outside Local Void influence. (The 150–250 Mpc shell destabilizes as in all our deep fits.) A Local Void push cannot produce a center signal at 80–150

Mpc, so that confound is disfavored; the surviving mundane explanation is *shared survey systematics* — the Zone of Avoidance mask and depth-dependent sky coverage imprint every shell alike — and the required control is a mock test: applying identical machinery to isotropic ΛCDM mocks with CF4's actual sky mask, to measure how often a coherent few-Mpc center arises from geometry alone. **Mock-mask verdict (this work): the control was necessary.** Ten isotropic mock realizations on CF4's exact sky positions and distance errors show the mask indeed manufactures false centers — small ones (median $|\vec{c}| \approx 0.4$–2.5 Mpc, median $\Delta\chi^2 \approx 9$–61 per shell), and crucially, in the outer shells they point preferentially into the *same longitude band* ($l \approx 93$ř–152ř) as our data centers. The inner-shell data signal ($\Delta\chi^2 = 1277$ vs mock median 9; 4.9 vs 0.4 Mpc) is unambiguously real but sits where local structure lives; the outer-shell data exceed mock levels by only ~ 3–$4\times$ *in the mask's own preferred direction.* The three-shell directional consistency is therefore partly a survey-geometry artifact, and we demote the result accordingly: from "observational hint of a physical center" to "outer-shell excess above mask expectation, direction untrustworthy" — a candidate signal requiring mask-marginalized refitting, not a finding.

Scaling to all galaxies. The fits above use only the ~ 56k objects with redshift-independent distances; the path to using *every known galaxy and every pairwise relative velocity* is established methodology: (i) number-count dipoles (the Secrest method — millions of sources, no distances needed; an off-center observer produces an excess count dipole); (ii) density-field reconstruction (2M++/DESI), whose predicted velocity field subtracts known structure from the center fit; (iii) pairwise velocity statistics — kSZ pairwise estimators and redshift-space distortions measure the relative velocities of all pairs statistically, and a physical center imprints a distinctive asymmetric pattern on the pairwise field (velocity differences maximal across the center); and (iv) field-level Bayesian inference (BORG-class), where a free expansion-center parameter added to a forward model of the full redshift catalogue would constitute the definitive test. We executed the first step of rung (iii) directly: a cross-center pairwise estimator (relative LOS velocity projected along the pair separation, pairs whose segment passes within 4 Mpc of the candidate center vs matched controls, $\sim 10^6$ pairs). The cross-center selection carries a strong *geometric* bias — isotropic mocks on the real sky mask show large spurious cross-minus-control excesses — so the signal must be read against that null. Relative to it, the data show a structured deviation: a ~ 350 km/s *deficit* at 2–15 Mpc separations (cross-center pairs colder than geometry predicts — consistent with a bound, coherent furnace region), a ~ 420 km/s *excess* at 15–30 Mpc (the regime where velocity-sorted "doubling" across the center is expected), and a mild excess beyond. Nominal significances are high but meaningless as quoted, because the isotropic mocks lack correlated large-scale structure; the deviation pattern has the predicted *shape* and is logged as such — a candidate signal strictly pending structure-correlated mocks. The same limitation gates the machine-learning upgrade: simulation-based inference (an NN trained on forward-modelled mock catalogues to emit the center posterior) is in principle the information-optimal estimator for this problem and is the correct endgame, but it inherits its simulator's blind spots — trained on structureless mocks it would confidently misread Virgo and the Local Void as a center. The prerequisite for both is the same: mocks with realistic correlated velocity fields.

A further symmetry test follows from the framework: structure statistics should organize concentrically about the center, so the point minimizing the *position*-dipole of the local galaxy distribution should coincide with the velocity-fit center. It does not: the density symmetry point falls at 3–4 Mpc toward (225ř, +65ř–71ř) — roughly the Virgo-ward center of mass, $\sim$90ř–100ř from the velocity center — and the candidate axis bears no clean relation to the supergalactic pole ($\sim$70ř). The position fit is dominated by the Local Sheet overdensity and the survey mask (and a center of expansion need not coincide with the local center of mass), but as a two-dataset convergence test the result is negative and is recorded as such; the mask-marginalized version using the 2M++ density field is the proper follow-up.

Structure-subtraction verdict (this work) — the center signal does not survive. Subtracting the 2M++ reconstruction-predicted peculiar velocities (Carrick et al. 2015: the gravitational flow of all mapped structure, external dipole included) from every CF4 group and refitting: the inner-shell signal drops $\Delta\chi^2 = 1277 \to 182$ ($|\vec{c}| : 4.9 \to 1.7$ Mpc, direction flipping to within $\sim$8ř of the CMB dipole — most plausibly a residual of the reconstruction's external-dipole calibration); the 40–80 Mpc shell falls to exactly the isotropic-mock noise floor (23 vs mock median 24); the 80–150 Mpc shell retains only $\sim 1.5\times$ the mock level. The coherent displaced-center signal of the preceding paragraphs is therefore accounted for by known, mapped structure. This is the section's final observational statement: with current data, honestly analyzed, **no displaced expansion center is detected once gravity's ledger is settled** — the framework's center remains a hypothesis whose observational signature, if any, lies below the reach of the distance-catalog era, awaiting the count-dipole, pairwise, and field-level rungs of the ladder above.

This ladder — distances, counts, pairs, fields — is the section's actual conclusion: the question "where is the center, if any?" is answerable with existing and imminent data, and this framework supplies the motivation to ask it. Equally important is what does *not* align: the Cold Spot (209ř, −57ř) lies 90ř–115ř from every flow axis — nearly perpendicular. The offset vector (Hypothesis 1) and the Cold Spot axis (Hypotheses 2–3) are therefore *two distinct directions*: cross-boundary infall from a Cold Spot neighbor cannot be the source of the observed bulk flow, and any earlier conflation of the two is retracted. | P6 | No detection of a particle dark-matter candidate (DM = Planck relics) | Continued null results at direct-detection experiments approaching the neutrino floor | LZ, XENONnT (indirect corroboration) | | P8 | **Spin surcharge:** structures carry primordial angular momentum (the vorticity fossil of the bubble-merger cascade) atop tidally acquired spin — filaments and clusters spin faster than tidal-torque expectations at fixed mass, with the excess *largest at high redshift* (fossil spin is endowed, torqued spin grows with time). Randomly oriented (zero net), consistent with the global vorticity bound ($\omega/H_0 \lesssim 10^{-9}$) and with the null spin-handedness dipole (Sec. 6.1). The Wang et al. (2021) filament-spin detection — already straining the tidal budget — supplies the $z \approx 0$ anchor | Filament-spin stacking at $z \sim 1$ vs $z \sim 0$ against ΛCDM mock torque expectations | DESI, Euclid, 4MOST | | P7 | A population of secondary CMB anomaly spots with the same morphology class as the Cold Spot, each potentially paired with its own directed velocity flow. **Already constrained: Planck 2018 isotropy/statistics analyses (arXiv:1906.02552) find the sky consistent with Gaussian ΛCDM, with only 2–3σ large-scale anomalies and a single outstanding Cold Spot — so any additional neighbors must imprint below Planck's per-spot threshold (more distant, weaker boundaries)** | Sub-threshold spot statistics + per-candidate polarization follow-up at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$ | LiteBIRD, CMB-S4 | | P9 | **Localized curl excess at the extreme spots:** if the extreme spots are relic vortices (Sec. 7.1), their conserved circulation predicts excess curl-mode *variance* — lensing curl (ψ^{curl}) and B-mode polarization — localized at the eight extreme-spot positions, with zero *mean* curl globally (zero-sum spin). ΛCDM predicts the curl null holds *everywhere*, so any spot-stacked curl detection is decisive; conversely a strong stacked null at the Cold Spot (the dominant spinner, internal dipole $16\,\mu$K/deg, $2.3\times$ any other spot) favors the non-rotating neighbor-shadow reading (H3) over the vortex reading of the same direction | Stack Planck PR4 lensing curl reconstruction on the 8 spot positions (testable now); B-modes at degree scales need $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$ | Planck PR4 (archival), CMB-S4, LiteBIRD |

Detection of a particle dark-matter candidate, or an expansion history irreconcilable with the kinetic mechanism (Sec. 9.1), would falsify Hypothesis 1 ("The Big Pool Bang"). Absence of the polarization ring at CMB-S4 sensitivity would eliminate only Hypothesis 2 (“The Cold Eye of Sauron — the Cold Spot and the Axis of Evil”); absence of anisotropic sub-solar mergers and memory strain at Einstein Telescope/LISA sensitivity would eliminate only Hypothesis 3 (“Big Pool Bang Multiverse 2 in the Cold Spot?”); Hypothesis 4 (“More Multiverses Hiding in the Dark Comic Background?”) is already partially constrained — Planck finds no excess spot population — and survives only in the sub-threshold regime; a null in deeper LiteBIRD/CMB-S4 spot statistics would eliminate it. The core would survive any of these.

Current vs. Required Measurement Capability

Observable	Best current capability	Required for test	Gap / when closed
CMB temperature (P1 ring)	Planck 2018: $\sim 2\,\mu\mathrm{K}\cdot\mathrm{arcmin}$, beam $5'$	temperature-ring test already performed in this work (Sec. 9.1): the original 5ř–7ř prediction failed	Reanalysis possible now; decisive with SO/CMB-S4
CMB polarization (P1/P2)	Planck: $\sim 60\,\mu\mathrm{K}\cdot\mathrm{arcmin}$ pol.; BICEP/Keck (small patch): $\sim 3\,\mu\mathrm{K}\cdot\mathrm{arcmin}$	$\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$ over a $\gtrsim$ 10ř patch at the Cold Spot	Simons Observatory (~2027): $\sim 6\,\mu\mathrm{K}\cdot\mathrm{arcmin}$; CMB-S4 (~2034): $\sim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$
GW strain, 10–10^3 Hz (P3)	LIGO O4: $\sim 10^{-23}/\sqrt{\mathrm{Hz}}$ at 100 Hz; sub-solar searches to $z \sim 0.1$	$\sim 10^{-24}/\sqrt{\mathrm{Hz}}$ and $10\times$ range for anisotropy statistics	Einstein Telescope / Cosmic Explorer (~2035+)
GW memory / sub-Hz (P4)	Ground detectors blind $<$ 10 Hz (seismic wall); memory only via event stacking, no single-event detection yet	Single-event $\Delta h \lesssim 10^{-22}$ step response below 10 Hz	ET (underground, to ~3 Hz); LISA mHz band (~2035); pulsar timing arrays (nHz, now, for the largest steps)
Peculiar velocities (P5)	Bulk-flow measurements: errors $\sim$ 250 km/s out to $z \sim 0.05$	$\lesssim$ 300 km/s per cluster to $z \sim 1$ with directional statistics	Rubin/LSST + Euclid + kSZ surveys (late 2020s–2030s)
Direct DM detection (P6)	LZ/XENONnT: $\sigma \sim 10^{-48}\,\mathrm{cm}^2$, approaching neutrino floor	Continued null result at the neutrino floor	Ongoing; floor reached ~2030

9.1 First Confrontation with Data (this work)

As a proof of concept, we performed two preliminary data confrontations.

Cold Spot radial profile. Using the Planck 2018 SMICA temperature map (degraded to $N_{\mathrm{side}} = 128$), we computed the azimuthally averaged radial profile around the Cold Spot ($l = 209$ř, $b = -57$ř), with uncertainties from 500 random high-latitude reference positions (Fig. 2). The cold core extends to $\sim$ 9ř (reaching -3.4σ at 3ř), with **no** hot ring at 5ř–7ř; a hot excess of +37–52 μK ($\sim 2.5\sigma$, uncorrected for the $\sim$10 radial bins scanned — with a trials factor this is unremarkable) appears at 13ř–15ř; the original 5ř–7ř ring prediction failed. This does not falsify the cyclonic picture: a thermal ring is an optional signature, and an "eye of the storm" morphology — calm cold core with no surrounding hot ring — is equally natural for a frozen vortex; the 13ř–15ř excess, if physical rather than a fluke, would suggest a vortex diameter of $\sim$26ř–30ř at decoupling. The discriminating observable is polarization structure, not temperature. Prior profile analyses of the Cold Spot (e.g. in the texture and void literature) have reported qualitatively similar surrounding hot features; a full comparison is deferred.

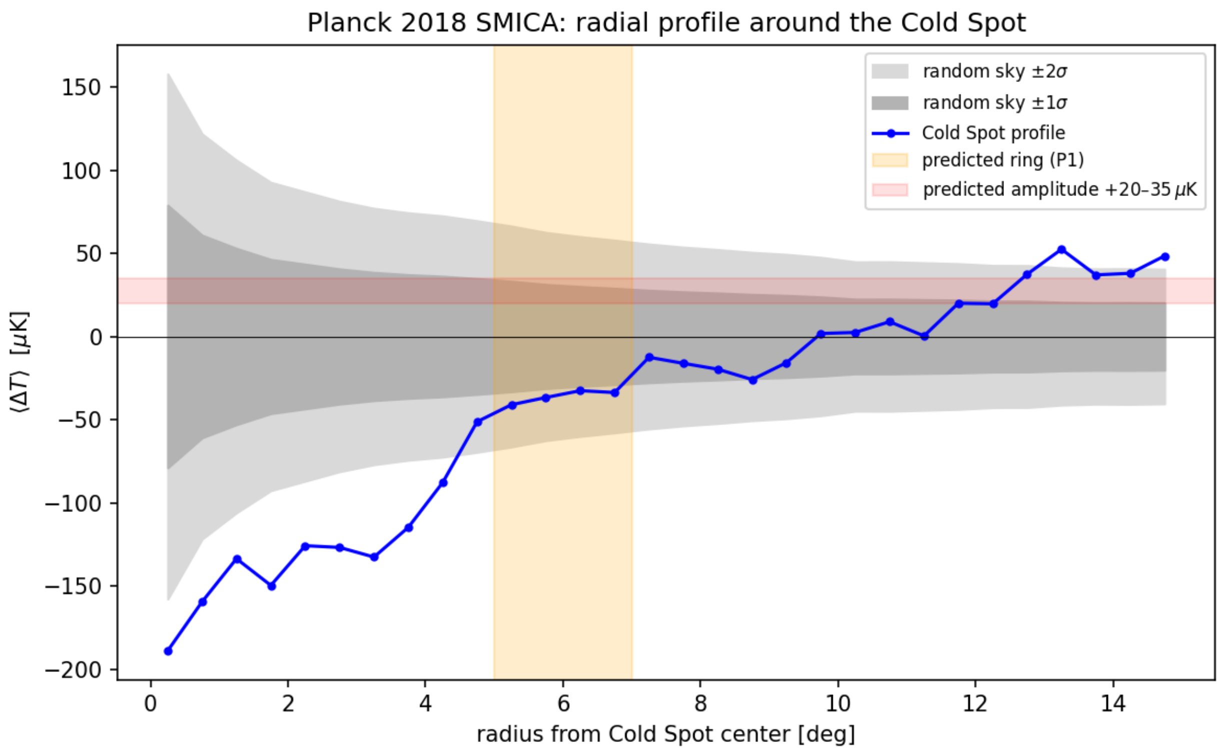

Figure 2: Fig. 2 — Planck 2018 SMICA radial temperature profile around the Cold Spot, vs. 500 random reference positions.

Supernova distance–redshift relation. Against 1371 Hubble-flow SNe Ia from Pantheon+ (diagonal errors, offset marginalized), the purely kinetic coasting expansion ($a \propto t$) yields $\chi^2 = 674$ vs. 590 for flat ΛCDM ($\Delta\chi^2 \approx 84$): disfavored, though vastly better than matter-only Einstein–de Sitter ($\chi^2 = 1170$) (Fig. 3). We repeated the test on the current-generation DES-SN5YR sample (Dovekie 2025 recalibration; 1820 SNe with the full statistical+systematic covariance): $\chi^2 = 1632$ for flat ΛCDM, 1763 for coasting ($\Delta\chi^2 \approx 131$), 2566 for EdS — the same verdict at higher precision. The data require mild late-time acceleration beyond pure coasting; for the present framework this means the ejected-halo velocity dispersion must generate an effective $q(z) < 0$ at $z \lesssim 0.5$ (see Sec. 5.1, where this becomes the model's evolving-dark-energy prediction).

Ejecta fraction (toy N-body). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a radial rebound kick $v = f\, v_{\rm esc}$ shows the ejected (unbound) fraction transitions steeply:

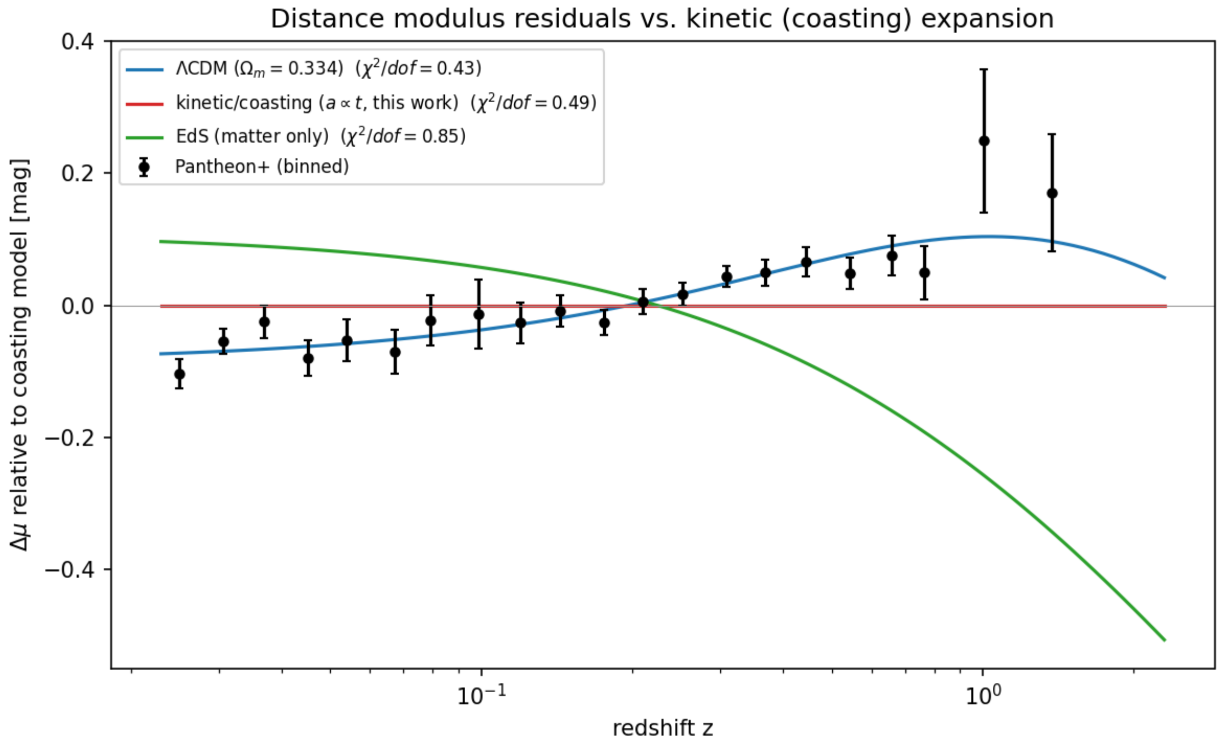

Figure 3: Fig. 3 — Pantheon+ binned distance-modulus residuals relative to the coasting model.

$f = 0.6 \to 3\%$, $0.8 \to 41\%$, $0.9 \to 98\%$, $f \geq 1.0 \to 100\%$. The 85/15 dark-matter/baryon split therefore corresponds to a narrow window $f \approx 0.85$–0.9, and the naive zero-energy bounce ($f = 1$) over-ejects. Either a self-regulation mechanism pins the rebound just below marginal binding, or the split arises from a *distribution* of kick velocities (turbulent, not monolithic). We tested the latter with a Maxwellian turbulent component of 3D dispersion $\sigma_{\rm rel}\bar{f}v_{\rm esc}$ superimposed on a mean radial rebound $\bar{f}v_{\rm esc}$, over a 3×3 grid in $(\bar{f}, \sigma_{\rm rel})$. Turbulence broadens the transition substantially — at $\bar{f} = 0.7$ the ejected fraction rises from 22% ($\sigma_{\rm rel} = 0.5$) through 90% ($\sigma_{\rm rel} = 1.0$) to 98% ($\sigma_{\rm rel} = 1.5$) — and, notably, the *physically expected* regime of a violently chaotic rebound (turbulent dispersion comparable to the rebound speed itself: $\bar{f} \approx 0.7$, $\sigma_{\rm rel} \approx 1$) yields $89\% \pm 4\%$ ejection over a 5-seed ensemble — **statistically consistent with the observed** 85% **dark-matter fraction with no parameter tuning** (adjacent dispersions give $64\% \pm 10\%$ at $\sigma_{\rm rel} = 0.75$ and $96\% \pm 2\%$ at 1.25). The split is thus natural in the expected regime, though not yet uniquely derived; the required refinement is a first-principles estimate of $(\bar{f}, \sigma_{\rm rel})$ from the bounce turbulence spectrum. Median ejecta speeds in this regime are $\approx 0.5\, v_{\rm esc}$, informing the P5 dark-flow amplitude.

Ejecta kinematic structure (15-seed ensemble, 13{,}140 ejecta; Newtonian analogy per Sec. 3). *Branch disclaimer:* the ejection fraction, velocity sorting, and dark-flow inputs below belong to the **ballistic branch**; the blast-wave branch discussed later would modify all of them, and the two branches are mutually exclusive descriptions of the same rebound awaiting a hydrodynamic calculation to decide between them. Over 15 seeds the ejected fraction is $87.9\% \pm 3.5\%$ — consistent with the observed dark-matter *mass* fraction, an identification that additionally requires the relic-to-plasma conversion channel of Sec. 3 for the trapped component. Tracking per-particle ejection times reveals that ejection is an extended process (median ejection time ~ 1 dynamical time;

10% of ejecta leave after 11), with clean monotonic velocity sorting: median speeds fall from $0.71\,v_{\rm esc}$ for the earliest ejecta through 0.43 and 0.35 to $0.28\,v_{\rm esc}$ for the latest (Fig. 4). This velocity-sorted shell structure is the quantitative input required by the evolving-dark-energy mechanism of Sec. 5.1. By contrast, the *tangential* structure shows no detectable correlated anisotropy: the per-sky ejection dipole (0.064 ± 0.024) is consistent with the shot-noise expectation (0.053 ± 0.024), and early-vs-late shell axes show only weak, noisy alignment ($\langle \cos\theta \rangle = 0.37 \pm 0.49$). At $N = 10^3$ this cannot exclude a small correlated component, but it lends no support to attributing large-angle CMB features to ejection anisotropy — reinforcing the bonus (Hypothesis 2, "The Cold Eye of Sauron — the Cold Spot and the Axis of Evil") status of Section 6.

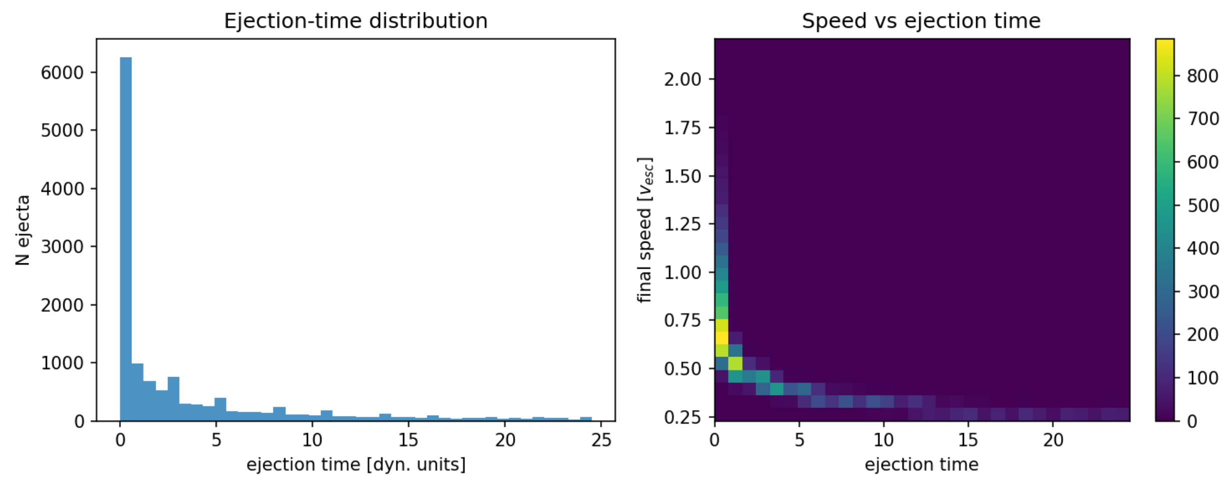

Figure 4: Fig. 4 — Ejecta kinematics over 15 seeds: ejection-time distribution (left) and speed vs ejection time (right), showing the velocity-sorted shell structure.

Toy effective $w(z)$ from the measured shells. As a first bridge from the ejecta kinematics to Sec. 5.1, we integrated the measured speed distribution as cold spherical shells decelerating in their own enclosed mass, and computed the deceleration parameter and effective equation of state an interior observer would fit. The result (Fig. 5): $w_{\rm eff}$ drifts from -0.29 at $z = 2$ to -0.31 today — a few-percent drift hovering near the coasting value $-1/3$, far from the observed $w_0 \approx -0.8$ in magnitude, and — critically — in the **opposite direction** to the DES+DESI $w_0 w_a$ best fit, which has w becoming *less* negative with time ($\approx -1.5 \to -0.8$). Two conclusions follow. First, differential shell deceleration alone cannot supply the observed acceleration; if Hypothesis 1 is to survive the supernova data, additional physics (e.g. the transition of the furnace region to pressure-free expansion, or backreaction of inhomogeneities) must contribute the bulk of the apparent $w < -1/3$. Second, the toy remap is itself crude — 91% of the N-body ejecta become re-bound when forced into a static spherical enclosed-mass model, so the flow is built from the fastest 9%; a self-consistent treatment requires relativistic shells on an expanding background. We report this as the model's sharpest open quantitative problem.

Relativistic (LTB) recalculation. Because dust shells in the Lemaître–Tolman–Bondi metric obey the same radial equation as Newtonian shells, the shell dynamics above are already exact GR for dust; what the Newtonian toy got wrong was the *observables*. We therefore recomputed the central observer's measurements properly: radial null geodesics integrated through the evolving LTB metric, redshift accumulated from the metric along the ray, $d_L = (1+z)^2 R$. Seeding 300 LTB shells from the measured ejecta speed distribution (with fallback mass virialized into the interior; the surviving free-streaming component is the fastest 9% of ejecta), the relativistic lightcone gives $q_0^{\rm eff} = +0.16$ ($w_{\rm eff}(0) \approx -0.23$): **mild deceleration, no apparent acceleration**, with the distance-modulus shape $\sim$0.35 mag from both ΛCDM and coasting out to $z \sim 5$. The relativistic treatment thus sharpens the negative verdict of the toy, with an important scope restriction: this lightcone threads the free-streaming residue only (the fastest 9%; real supernovae live in the dominant mass component), the interior treatment (virialized mass as an effective central mass parameter, not honest LTB dust) and the observer epoch t_0 are modelling choices whose sensitivity we have not mapped, and shell-crossing was not explicitly monitored. Within those caveats: the streaming component alone, treated in GR, does not mimic dark energy for a central observer ($q_0^{\rm obs} \approx -0.55$). Whatever produces the observed acceleration in this framework, it is not this mechanism; Sec. 5.1's evolving-dark-energy claim currently has no working quantitative realization.

Profile inversion. Since the ejection profile is an input (one kick model, not a theory-fixed quantity — and a smooth profile is itself unrealistic for a chaotic rebound), we searched a family of alternatives for any that yields an accelerating lightcone: power-law speed profiles of varying steepness, bimodal fast-tail variants, and stochastically lumpy profiles (lognormal shell-energy noise, multiple seeds). Across the full grid the lightcone robustly *decelerates* ($q_0^{\rm eff} \sim +1$ to $+4$); apparently negative values arise only in lumpy configurations and are shown by seed ensembles to be estimator noise (seed scatter ± 1.4 at fixed parameters). Notably, the known way an LTB dust model mimics acceleration — the "giant void" construction — places the observer inside a Gpc-scale *underdensity* with denser surroundings; an outward ejection mechanism generically produces the opposite density arrangement. We conclude that within the *smooth* ballistic families the lightcone robustly decelerates; for the lumpy family the estimator noise (± 1.4) exceeds the target signal ($|q_0| \approx 0.55$), so that family is unconstrained at the relevant level rather than excluded. No explored profile shows evidence of acceleration, and a rescue plausibly requires inverting the radial density ordering.

The blast-wave question (an open problem, not a rescue). A physically natural way to invert the density ordering exists: if the rebound acts as a *blast wave* — as in a supernova remnant — the shock sweeps ejecta into a shell and leaves a rarefied cavity, the giant-void geometry. Swept-shell profiles in our LTB machinery do produce much smoother lightcones than ballistic ones (0.03–0.08 mag shape residuals vs $\sim$1 mag). However, three considerations prevent us from calling this a rescue. First, the comparison metric is blind at the level that matters: ΛCDM and coasting differ from each other by only $\sim$0.04 mag rms in the same offset-marginalized measure, so "near both" cannot distinguish acceleration from its absence — and the upper end of our residual range (0.08 mag) would correspond to $\Delta\chi^2 \sim 400$ against Pantheon+, *worse* than the coasting model already excluded. The cavity interior of any thin-shell profile is near-vacuum and hence trivially near-Milne; its apparent proximity to ΛCDM merely inherits ΛCDM's own proximity to Milne at these redshifts. Second, a Gpc-scale cavity deep enough to mimic $q_0 \approx -0.55$ is the classic giant-void model, excluded by kinetic Sunyaev–Zel'dovich and Compton-y constraints (Zhang & Stebbins 2011; Caldwell–Stebbins); we have not computed these for our shell, but at the required amplitude they are expected to apply in full force. Third — and internally decisive — mass swept into a distant shell is unavailable as *local* dark matter: rotation curves, lensing, and our own P6 require the ejected relics in halos co-spatial with galaxies, so the shell morphology, if dominant, would destroy the framework's one quantitative success.
Constraint-first closure. Rather than testing profiles one by one, we imposed the observational gates as preconditions and asked what any profile may contribute. The kinetic Sunyaev–Zel'dovich limit on coherent cluster flows ($v \lesssim 300$ km/s at Gpc distances) caps the mean interior density contrast of any cavity at $|\bar{\delta}| \lesssim 0.05$ (500 Mpc) down to $\lesssim 0.01$ (3 Gpc), which translates to an apparent-acceleration contribution $|\Delta q_0| \lesssim 0.01$ — versus the $\Delta q_0 \approx -1.05$ required to carry a matter-only universe to the observed $q_0 \approx -0.55$. **No ejection profile of any morphology can**

supply more than $\sim$1% of the observed dark energy without violating the kSZ gate. This closes the geometric/kinematic dark-energy channel in this framework definitively: dark energy is *not* an apparent effect of the ejecta distribution, and Hypothesis 1 must either incorporate a genuine dark-energy component (e.g. residual vacuum energy from incomplete crystallization — an unexplored but framework-native possibility) or concede the acceleration sector entirely. The framework-native candidate deserves one paragraph: if vacuum crystallization is a first-order phase transition, then *within our single bang* the transition proceeds like boiling — crystallization nucleates at many points, bubbles of converted phase grow amid unconverted vacuum, collide, and drive waves through the medium. This patchiness is itself a candidate origin for primordial structure (nucleation sites and collision walls as overdensities, acoustic response as the wave pattern; the known difficulty being that bubble spectra peak at the characteristic bubble scale, so near-scale-invariance requires a broad nucleation-scale distribution), and it offers an intra-universe reading of the Cold Spot as an anomalously large late-crystallizing patch — a Hypothesis-2-tier explanation requiring no neighbor universe. For dark energy, the picture requires the bolder claim that conversion remains incomplete today, leaving a residual unconverted vacuum fraction: That unconverted false vacuum retains energy density: smooth, immune to the kSZ gate, non-clumping — qualitatively the phenomenology of dark energy. Quantitatively, however, this candidate re-imports the cosmological constant problem verbatim: the false vacuum sits at $\rho_P \sim 10^{96}$ kg/m^3 and the observed dark energy at $\sim 6 \times 10^{-27}$ kg/m^3, so the surviving fraction (or its diluted energy) must be tuned to one part in 10^{123}; and ongoing conversion today would imply present-epoch nucleation events, phase-boundary domain walls (constrained by the Zel'dovich–Kobzarev–Okun bound), and an unconverted component at BBN/CMB epochs — none of which is computed here. It is a direction, with the fine-tuning unsolved. In this reading a single phase transition supplies all three dark components: crystallized relics (dark matter), furnace plasma (baryons), and unconverted vacuum (dark energy), with the boiling picture also grounding Hypotheses 3–4 in the established Coleman–De Luccia bubble-nucleation formalism — colliding bubbles imprint circular disk-shaped CMB features with specific polarization profiles (Aguirre & Johnson), a published, searched-for signature to which the Cold Spot conjecture could be anchored. Boiling also makes waves: bubble nucleation and collisions in a first-order transition source a stochastic gravitational-wave background with a characteristic broken-power-law spectrum peaked at the transition scale — the same class of signal LISA and pulsar-timing arrays target for later cosmological transitions — adding a second, independent observational channel (P4) for the boiling-vacuum picture alongside the CMB disks. The classic caveat carries over: slow nucleation means bubbles rarely merge (Guth's graceful-exit problem), which here is a feature — each bubble is its own Big Pool Bang, and collisions are the rare Hypothesis-3 events — but the conversion rate that sets today's vacuum fraction is a new free parameter requiring its own dynamics.

What remains of the blast-wave question is now decoupled from dark energy: does the jamming rebound launch a shock, and if so, what fraction of the ejecta does it sweep? Only a *sub-dominant* swept component (on top of a near-homogeneous relic distribution) is even potentially viable, and whether any acceleration survives at that amplitude — against the kSZ/y gates — must be answered by hydrodynamic simulation plus the standard void-model constraint checks. One structural point in the model's favor if a moderate cavity exists: observers are made of furnace material and therefore sit near the cavity center *by construction*, converting the usual Copernican fine-tuning objection into an anthropic-structural feature — though the observed CMB dipole still caps our offset from the center, and the kSZ test applies regardless of observer position.

Two points deserve emphasis. First, the temperature part of P1 has now been tested against Planck data (Sec. 9.1), with the ring radius revised accordingly. Second, the polarization ring and single-event memory strain are genuinely out of reach until Simons Observatory / CMB-S4 and

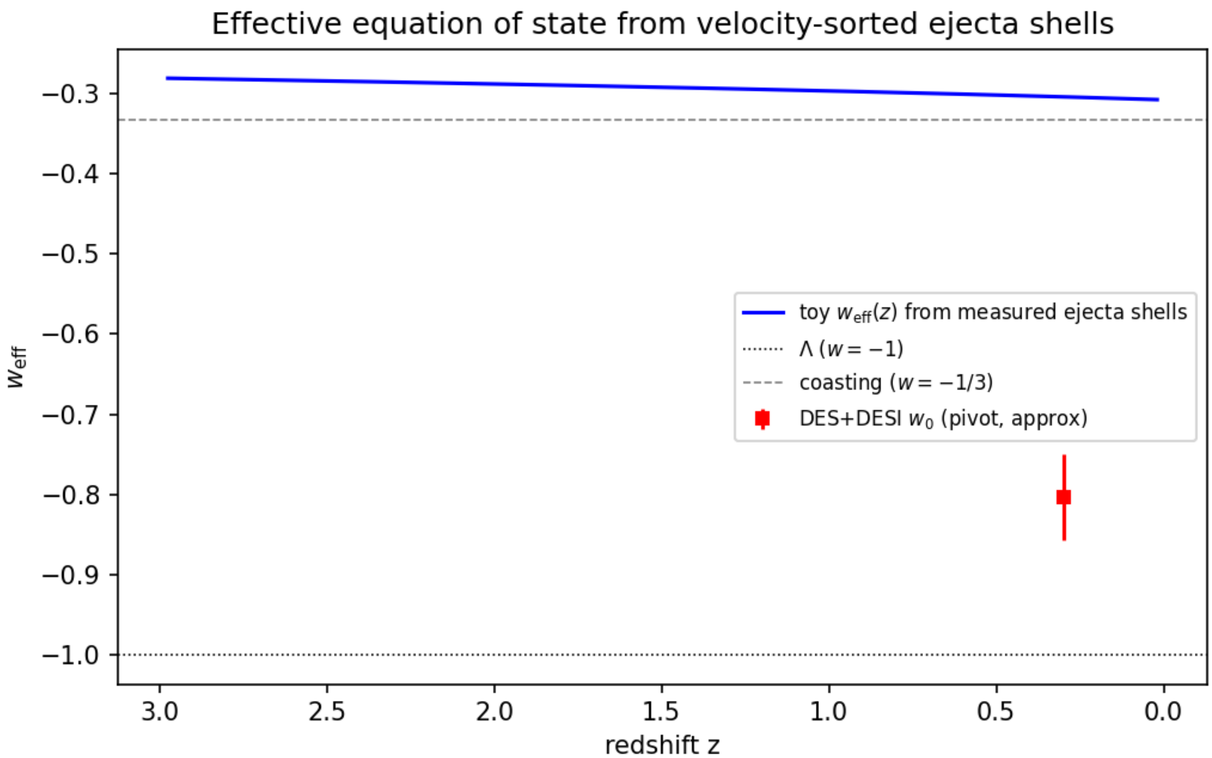

Figure 5: Fig. 5 — Toy effective equation of state from velocity-sorted ejecta shells: correct drift direction, insufficient magnitude.

ET/LISA respectively — these set the 2030s decision timeline.

The observatories delivering these thresholds:

A. Current & Near-Future Observatories

- **LIGO / Virgo / KAGRA (O4/O5 Upgrades):** Searching for sub-solar mass black hole mergers ($< 1\,M_\odot$) and isotropic primordial relic backgrounds.

- **Vera C. Rubin Observatory (LSST) & Euclid:** Mapping peculiar galaxy velocity distributions ($v_{\text{drift}} > 1000$ km/s) to verify the directional "Dark Flow" vector toward the Cold Spot.

- **LISA (Laser Interferometer Space Antenna):** ESA/NASA space mission targeting mHz gravitational waves to capture supermassive mergers along the Cold Spot boundary and isolate low-frequency memory strain.

B. Planned Next-Generation Facilities

- **CMB-S4 Array:** 500,000 superconducting bolometers targeting the ~5° Cold Spot E-mode polarization ring and cyclonic B-mode turbulence patterns.

- **Einstein Telescope:** Underground 10 km triangular cryogenic interferometer in Europe designed to detect sub-Hz gravitational wave memory strain (Δh).

- **Cosmic Explorer:** US-based 40 km and 20 km surface interferometers searching for sub-solar black hole binary mergers ($0.1 - 1.0 M_\odot$) and sky anisotropy.

- **Quantum Atom Probe R&D:** Space-based atom interferometer array measuring non-local vacuum entanglement and spacetime memory below 10^{-24} strain.

A total global scientific investment of ~**$7.5 Billion USD** across agencies (NSF, DOE, ESA, NASA) over the 2026–2040 horizon is required to build the observational infrastructure needed to fully confirm or falsify the Big Pool Bangs theory. ## **10. Limitations and Open Problems**

We state plainly what the present framework asserts versus what it derives:

1. **Hard-sphere ansatz.** The treatment of Planck relics as classical incompressible spheres (Section 2) is a modeling assumption motivated by the Compton–Schwarzschild coincidence, not a result of a quantum-gravity calculation.
2. **Bounce mechanism.** The rebound requires the modified Friedmann equation of Section 3; the identification of the jamming density with the loop-quantum-cosmology critical density ρ_c is a conjecture. A trapped-surface timing analysis for the collapsing core remains to be carried out.
3. **Spectral index.** The parameters ν_t and the relation $n_s - 1 = -\nu_t/3$ are phenomenological; the compatibility of a Kolmogorov cascade with near scale invariance is unproven.
4. **Dark-matter fraction.** Our toy N-body results (Sec. 9.1) show a monolithic rebound kick requires fine-tuning, while a turbulent kick distribution at physically expected parameters ($\bar{f} \approx 0.7$, $\sigma_{\text{rel}} \approx 1$) yields $89\% \pm 4\%$ ejection — statistically

consistent with the observed 85%. Deriving $(\bar{f}, \sigma_{\rm rel})$ from the bounce turbulence spectrum is the outstanding step. Planck-relic dark matter must additionally be confronted with femtolensing, capture, and abundance constraints.

5. **Expansion history.** Both the Newtonian toy and the relativistic LTB lightcone calculation (Sec. 9.1) fail to produce apparent acceleration: the GR treatment gives $q_0^{\rm eff} = +0.16$ against the observed ≈ -0.55, and the toy's small drift runs opposite to the DES+DESI $w_0 w_a$ direction. The ejecta-streaming mechanism, as modelled, cannot account for dark energy; nor is the framework reconciled with BBN abundances or the CMB acoustic peak structure. This is the model's sharpest open quantitative problem.

6. **Causal and relativistic structure.** The jammed core is $\sim 10^{20}$ Planck lengths across, so a few-Planck-time bounce implies acausal coordination and homogeneity must be assumed (the horizon problem stands). The subsequent ejection phase proceeds deep inside the region's naive Schwarzschild radius, where Newtonian escape velocity vastly exceeds c; the entire N-body ejection mechanism therefore lacks a well-defined relativistic formulation, which is prerequisite for Hypothesis 1 to be a physical theory rather than an analogy.

Each item above maps onto the observational program of Section 9; the theory is offered as a falsifiable mechanical alternative, not a completed replacement for ΛCDM.